CHAPTER

5

Group Reporting IV

Consolidation under IFRS 10



LEARNING OBJECTIVES

After reading this chapter, you should be able to:

LO1 Understand the principles underlying the elimination of intragroup balances and transactions in consolidation;

LO2 Understand the rationale for consolidation adjustments to opening retained earnings;

LO3 Appreciate the significance of upstream versus downstream sales and the consequential impact on non-controlling interests;

LO4 Pass the appropriate consolidation adjustments with respect to unrealized profit or loss arising from intercompany transfers of inventory and fixed assets; and

LO5 Understand and know how to analytically determine consolidated retained earnings.

ELIMINATION OF INTRAGROUP TRANSACTIONS AND BALANCES

Since the existence of a group is motivated by the economic benefits to be gained from operating as one unit, there are likely to be significant operational and financial interdependencies among the entities within a group that may lead to intragroup transactions and balances. Intragroup transactions arising from operational and financial interdependencies include:

1. Selling inventory to, or buying inventory from, a group company;
2. Transfering long-lived assets (for example, fixed assets or investments) from one group company to another;
3. Rendering service to, or procuring services from, other group companies; and
4. Providing financing for working capital or long-term financing needs of group companies.

The transfer of assets within a group is rarely transacted at the carrying amounts of the assets as recorded in the books of the selling company. Business units within a group often operate competitively. If the transaction is done on an arm's-length basis, the transaction price is at fair value and a profit margin is often included in the transfer price. The difference between the transfer price and the book value or the carrying amount of the asset results in a profit or loss for the selling company. From a group's perspective, the profit or loss is unrealized or unearned until the asset is sold to a third party. The lag between the point of purchase and the point of resale often results in the inventory being recognized in the statement of financial position of the buying company at a marked-up price. While the profit is deemed to be earned from the legal entity's perspective in accordance with IFRS 15 *Revenue from Contracts with Customers*, both the profit of the selling company and the inventory of the buying company are overstated from the group's perspective. The control of ownership of the inventory has not been passed from the group to an external party, and inventory is measured at an amount that exceeds the original purchase price. From the group's perspective, the unrealized profit has to be eliminated, and the inventory has to be restated to the lower of original cost as transacted with third parties and net realizable value. The Group recognizes profit only when control of the asset is transferred to third parties or the asset is used, consumed or expensed off.

Intragroup transactions such as intragroup transfer of assets and intragroup financing give rise to *intragroup balances*.

Examples of intragroup balances include:

1. Amounts receivable from, or payable to, group companies arising from intragroup transactions such as the transfer of assets, provision of services, or charging of interest on intragroup borrowings;
2. Loans receivable from, or payable to, group companies; and
3. Dividends receivable from subsidiaries or dividends payable to parent company.

From an economic perspective, an entity is not able to transact with itself. Neither can an entity borrow from or lend monies to itself. Since consolidated financial statements are prepared for the group as a single economic entity, intragroup transactions and balances have to be eliminated in full.

IFRS 10 Appendix B paragraph B86(c) states that:

> Consolidated financial statements eliminate in full intragroup assets and liabilities, equity, income, expenses and cash flows relating to transactions between entities of the group (profits and losses resulting from intragroup transactions that are recognized in assets, such as inventory and fixed assets, are eliminated in full). Intragroup losses may indicate an impairment that requires recognition in the consolidated financial statements. IAS 12 *Income Taxes* applies to temporary differences that arise from the elimination of profits and losses resulting from intragroup transactions.

Consolidation adjustments are passed either to eliminate or correct, or reverse the original entries so that the final consolidated financial statements reflect transactions and balances with parties that are external to the group. The key

to understanding consolidation adjustments is to remember that these adjustments are in relation to the original entries that are passed in the legal entity financial statements of the individual group companies.

In practice, consolidation involves a three-step process.

1. Preparing consolidation adjusting entries to arrive at consolidated totals that reflect the effects of transactions of the economic entity with external third parties;
2. Preparing consolidation worksheets, which combine the legal entities' financial statements and the consolidation adjusting entries; and
3. Analyzing the final consolidated totals to independently substantiate the reported numbers.

In today's environment, the compilation of consolidation worksheets is performed through the use of readily available computerized software such as Microsoft Excel. More importantly, the accounting student should understand the process in preparing the consolidation adjusting entries and in analyzing the consolidated totals. These steps reflect understanding of the application of accounting principles at the economic entity level.

In this chapter, we explain the principles that govern the preparation of consolidation adjusting entries. We also show how final consolidated totals can be substantiated. The compilation of the consolidation worksheet and the preparing of consolidated financial statements is a mechanistic process that follows from the more critical stage of preparing consolidation adjusting entries. The final step of analyzing consolidated totals is also critical in ensuring an independent check of the reported numbers. Hence, the focus of this chapter is on the preparation of consolidation adjusting entries and the analysis of the consolidated totals.

Principles Governing Elimination

The principles governing elimination are as follows:

1. Outstanding balances due to, and from, companies within a group (that is, a group, as defined in IFRS 10 Appendix A, comprises a parent and its subsidiaries) are eliminated.
2. Transactions in the income statement between one group company and another are eliminated.
3. Profit or loss resulting from intragroup transactions that are included in the carrying amount of an asset at year-end must be eliminated in full. Elimination in full implies that the non-controlling interests' share of the profit or loss is eliminated as well. The consistent treatment between controlling and non-controlling interests with respect to elimination of unrealized profit or loss is in line with the entity theory of consolidation.
4. Tax effects on unrealized profit or loss included in the carrying amount of an asset such as inventory or fixed assets should also be adjusted in accordance with IAS 12 *Income Taxes*.
5. Associates (entities over whom an investor has "significant influence") are not included in the definition of "group". Thus, balances with associates are not eliminated. Unrealized profit or loss from transactions between an investor and its associates are eliminated to the extent of the investor's interest. The principles and processes relating to accounting for associates are elaborated in Chapter 6.

ELIMINATION OF REALIZED INTRAGROUP TRANSACTIONS

When the profit and loss effects are exactly offsetting for both the buying and selling company within the group, the reciprocal account items will be eliminated. Although there is a "natural" offsetting effect on net profit in the consolidated income statement, the individual items must be eliminated to avoid the overstatement of disclosed line items in the consolidated income statement. The following are examples of intragroup transactions that affect the income statement.

Transactions Relating to Interest

Generally, the interest expense should exactly offset interest income. For example, a parent lends $2,000,000 to a subsidiary at the rate of 5% per annum. Assuming a full year's interest charge, the elimination entry is as follows:

Dr Interest income (Parent)	100,000	
Cr Interest expense (Subsidiary)		100,000

While intragroup interest transactions should eliminate perfectly, there will be some situations when the interest income of the lending company does not offset the interest expense of the borrowing company. This situation arises when one group company lends to another to develop or construct a long-lived asset. IAS 23 *Borrowing Costs* requires the borrowing company to capitalize the interest during the development or construction period into the cost of the asset. As such, the cost of borrowing is not expensed off in the period of the borrowing but in the period when the asset is amortized, depreciated, consumed, or sold.

Let's consider the following example. Parent Co borrows from an unrelated bank $1,000,000 at the rate of 5% per annum and lends $1,000,000 to Subsidiary Co on 2 January 20x1 at an interest rate of 6% per annum for the purpose of constructing a warehouse. The warehouse was completed on 31 December 20x1 and has a useful life of 20 years from 1 January 20x2. Interest is paid on 31 December but the loan remains outstanding throughout the period. Ignore tax effects for simplicity.

During 20x1, Parent Co will record the following journal entries in its separate financial statements:

Dr Cash	1,000,000	
Cr Loan payable to bank		1,000,000
Dr Loan receivable from Subsidiary Co	1,000,000	
Cr Cash		1,000,000
Dr Interest expense	50,000	
Cr Cash		50,000
Dr Cash	60,000	
Cr Interest income		60,000

At the same time, Subsidiary Co will record the following journal entries and capitalize interest as asset cost:

Dr Cash	1,000,000	
Cr Loan payable to Parent Co		1,000,000
Dr Fixed assets in progress	60,000	
Cr Cash		60,000

At the economic entity level, the interest "income" does not exist as it is only an internal transaction. Instead, the external transaction is the borrowing from the bank to finance the construction of the fixed assets. Hence, the internal interest should be eliminated and the external interest should be capitalized into the cost of the asset in line with IAS

23 *Borrowing Costs*. The consolidation adjusting entries should, therefore, be as follows (these entries can be combined but shown here separately for ease of understanding):

31 Dec 20x1	Dr Interest income	60,000	
	Cr Fixed assets in progress		60,000
	To eliminate interest income and internal interest capitalized in fixed assets		

31 Dec 20x1	Dr Fixed assets in progress	50,000	
	Cr Interest expense		50,000
	To capitalize the external interest in self-constructed fixed assets		

The intercompany balances should also be eliminated:

31 Dec 20x1	Dr Loan payable to Parent Co	1,000,000	
	Cr Loan receivable from Subsidiary Co		1,000,000
	To eliminate the intercompany loan balances		

In 20x2, the construction of the fixed assets is completed and the asset will be depreciated over a 20-year period. The consolidated income statement should reflect the depreciation of the capitalized interest from external sources, not the interest that arises from the internal loan:

Depreciation from the group's perspective ($50,000/20)	$2,500
Depreciation from the legal entity's perspective ($60,000/20)	3,000
Excess depreciation to adjust on consolidation	$ 500

Consolidation adjusting entries for 20x2 is as follows:

31 Dec 20x2	Dr Interest income	60,000	
	Cr Interest expense		60,000
	Elimination of internal interest transaction		

31 Dec 20x2	Dr Accumulated depreciation	500	
	Cr Depreciation expense		500
	Adjustment of excess depreciation		

The group also needs to pass through an entry to "correct" the overstatement of opening retained earnings of Parent Co and the overcapitalization of interest in fixed assets as at 1 January 20x2. This entry essentially "re-enacts" the previous year's consolidation adjustments. In a later section, we explain in greater depth the significance of adjustments to opening retained earnings and the impact of the direction of the transfers (that is, from parent to subsidiary or subsidiary to parent). For now, understand the intuition that from the economic entity's perspective, the retained earnings of Parent Co and the fixed assets of Subsidiary Co are overstated by the interest transaction.

31 Dec 20x2	Dr Opening retained earnings	10,000	

	Cr Fixed assets		10,000

Adjustment of opening retained earnings and overstated fixed assets as at beginning of the year

Transactions Relating to Services Provided

As with interest, the provision of services and the enjoyment of the benefits of the services are simultaneous. Both parties should record the transaction in their separate financial statements in the period when the services are rendered. For example, Subsidiary A rents its warehouse to Subsidiary B for $200,000 a year. An elimination entry is needed to reverse the effects as recorded in the separate financial statements of each subsidiary, as follows:

Dr Rental income (Subsidiary A)		200,000	
Cr Rental expense (Subsidiary B)			200,000

As with interest transactions, there may be service transactions that are recognized in different periods because the group company that is the customer capitalizes part or all of the service costs in an asset. The capitalization situation may arise when the service provided creates or enhances an asset or extends its useful life. For example, if one group company provides architectural services to another, the fee income of the architect will be offset against the asset of the buying company.

For example, Parent Co provides architectural services to Subsidiary Co during 20x1 and charges a fee of $500,000 during that year, which is over and above directly attributable costs and employee benefits[1] incurred on the architectural project of $300,000 which were expensed off by Parent Co. The fees are capitalized into the building cost of Subsidiary Co. Depreciation begins on 1 January 20x2 over an estimated useful life of 20 years. Ignore tax effects for simplicity.

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In 20x1, the consolidation adjustment will eliminate the fee income against the fixed assets in progress:

31 Dec 20x1	Dr Fee income	500,000	
	Cr Fixed assets in progress		200,000
	Cr Architectural expense		300,000

To eliminate fee income and profit in fixed assets and to capitalize directly attributable costs

In 20x2, we need to pass consolidation adjustments to "re-enact" CJE1 to correct the opening retained earnings of Parent Co and the overstatement of fixed assets:

31 Dec 20x2	Dr Opening retained earnings	200,000	
	Cr Fixed assets		200,000

To eliminate fee income and capitalized cost in fixed assets as at 1 January 20x2

In 20x2, we also need to pass through consolidation adjustments to correct the excess depreciation of $10,000 (that is, $200,000 divided by 20 years) that arises from the internal fee capitalized in fixed assets:

31 Dec 20x2	Dr Accumulated depreciation	10,000	
	Cr Depreciation expense		10,000

To adjust excess depreciation on internal fee capitalized in fixed assets

Transfer of Inventories that are Resold to Third Parties Within the Same Financial Year

In such a situation, the intercompany sales by the selling company will exactly offset the intercompany cost of sales of the buying company. Any profit (or loss) of the selling company will offset the higher (or lower) cost of sales of the buying company. Even though there is no net impact on consolidated net income, the group has to pass a consolidation adjustment to eliminate the sale and cost of sale. Without this adjustment, the consolidated sales and consolidated cost of sales would include intragroup transactions. For example, Subsidiary C sells $500,000 worth of inventory to Subsidiary D, which then resells the inventory to third parties within the same financial year. Subsidiary C records a profit of $100,000 from the sale. The elimination entry is as follows:

Dr Sales (Subsidiary C)	500,000	
Cr Cost of sales (Subsidiary D)		500,000

In this example, the "marked-up" sales price of $500,000 exactly offsets the higher cost of sale of subsidiary D. From a group's perspective, this adjusting entry is necessary as the effects of the intragroup transfer should be fully eliminated, and the consolidated financial statements should show only the sale to third parties and the original cost of purchasing the inventory from third parties.

Dividend Transactions

A parent records dividend income when the parent's right to receive payment is established (IFRS 9 paragraph 5.7.1). When dividends are proposed but not declared, the parent should not recognize the dividend income as the right to receive payment is not established. A consolidation entry is necessary to reverse out the proposed dividends. Chapter 4 has explained and illustrated the elimination of dividends declared and dividend income.

ELIMINATION OF INTRAGROUP BALANCES

Prior to eliminating intragroup balances, a reconciliation must be carried out by both group companies to ensure that the reciprocal balances are in agreement. Frequently, reconciling items may be due to:

1. In-transit items that are recorded by one company but not by the other;
2. Errors and omissions; or
3. Dispute on the transaction.

All reconciling items must be resolved appropriately before intragroup balances can be eliminated. For example:

1. In-transit items should be either adjusted out or recognized in a manner that is consistent with the other party's treatment and should conform to sound accounting principles;
2. Errors and omissions should be corrected as appropriate; and
3. Disputed items should be either recognized by the disputing party or adjusted out by the party that recorded the items in its books.

An example of a reconciliation between two fellow subsidiaries is shown in Table 5.1.

TABLE 5.1
Reconciliation of intercompany balances between Subsidiaries A and B

Amount owing by B in A's books as at 31 December 20x5	$40,000
Less items in A's books but not in B's books:	

Invoice #1278 (5/11/05)	Disputed (Note 1)	(1,500)
Invoice #1309 (26/12/05)	Goods received on 29 December 20x5 (Note 2)	(3,200)
Invoice #2730 (31/12/05)	Goods received on 3 January 20x6 (Note 3)	(2,500)
Debit note #21 (31/12/05)	Repair for goods not under warranty (Note 4)	(300)
Less payment made in December 20x5 by B but not recorded by A:		
Cheque No. 1024 paid on 31 December 20x5 and uncleared as of 31 January 20x6 (Note 5)		(17,000)
Less items recorded by B but not by A:		
Defective goods returned on 1 December 20x5 by B but for which no credit note has been received (Note 6)		(2,000)
Amount owing to A in B's books as at 31 December 20x5		$13,500

Note 1: For purposes of consolidation, either Subsidiary A had to reverse the sales or Subsidiary B had to accrue for the invoice. However, both parties may reserve their rights to pursue a satisfactory resolution of the dispute, regardless of the accounting treatment.

Note 2: Since goods were received before the year-end, Subsidiary B had to record an adjusting entry as follows:

Dr Inventory	3,200	
Cr Payable to A		3,200

Note 3: Goods were received after year-end. If shipment had been made before the year-end, Subsidiary B would have recorded the transaction as follows:

Dr Goods-in-transit	2,500	
Cr Payable to A		2,500

Note 4: Since repairs were not covered under warranty, Subsidiary B had to make the following adjustment:

Dr Repair costs	300	
Cr Payable to A		300

Note 5: Follow-up action is required to ascertain the reason for the non-clearance of the cheque. The cheque was either lost in transit or unbanked by Subsidiary A. If lost, the cheque should be cancelled and reissued; an adjusting entry is required by Subsidiary B.

Dr Bank	17,000	
Cr Payable to A		17,000

Note 6: An adjusting entry by either Subsidiary A or Subsidiary B was necessary for purposes of agreement of intercompany balances, pending the satisfactory resolution of the matter.

After the reconciliations are properly carried out and followed up with appropriate entries in the separate financial statements, the elimination entries are simple when the balances agree. The examples of elimination entries are shown below:

Dr Intercompany payable (SFP)
  Cr Intercompany receivable (SFP)

Dr Dividend payable to parent (SFP)
  Cr Dividend receivable from subsidiary (SFP)

Dr Loan payable to parent (SFP)
  Cr Loan receivable from subsidiary (SFP)

ADJUSTMENT OF UNREALIZED PROFIT OR LOSS ARISING FROM INTERCOMPANY TRANSFERS

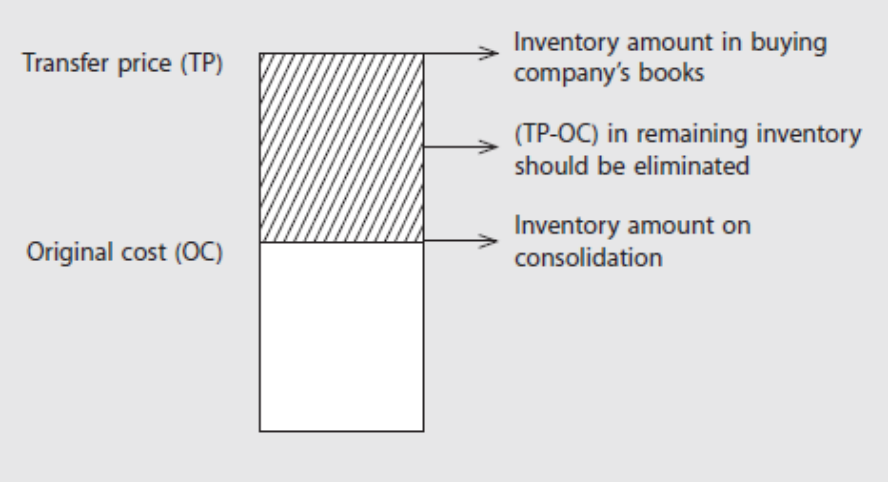
Profits and losses resulting from intragroup transactions that are recognized in assets, such as inventory and fixed assets, are eliminated in full (IFRS 10 App B: B86(c)). In other words, the carrying amounts of assets in the consolidated statement of financial position should not include any element of unrealized profit or loss arising from intercompany sales unless the loss is an impairment loss. In a later section, we will explain special considerations relating to transfers made at a loss.

Intragroup Transfers of Inventory and Fixed Assets

If the asset is *inventory*, it should be carried at the lower of cost and net realizable value. Cost is the original purchase price as transacted with an external party. The requirement in IAS 2 paragraph 9 to carry inventory at the lower of cost and net realizable value applies to both the economic (group) and legal entity. From the group's perspective, original cost is the exchange price when the goods were originally purchased from a third party. From the legal entity's perspective, original cost is the exchange price of the goods purchased from another group company. The other group company is an external party to the buying company from a legal perspective. The cost of inventory of the buying company would include the profit mark-up of the selling company.

A consolidation adjustment must be passed to eliminate the profit element included in the carrying amount of the inventory. When the inventory is eventually resold to a third party, the cost of sales of the inventory in the consolidated financial statements should be the original cost as transacted with unrelated third parties and not the transfer price invoiced by one group company to another. Figure 5.1 shows graphically the unrealized profit that should be eliminated from inventory.

FIGURE 5.1 Unrealized profit in inventory



If the asset is a fixed asset, it should be stated at its cost as determined from its original purchase transaction and deducted for any accumulated depreciation from the date of the original purchase to the current period. Adjustments in relation to fixed assets are discussed in the section "Special Considerations for Intercompany Transfers of Fixed Assets".

From a group's perspective, the depreciation of transferred fixed assets has to be calculated based on the original cost of the fixed asset and not the transfer price that is invoiced by the selling company to the buying company. If there is a revision of useful life or change in estimates, the changes for the group should be made with reference to the carrying amount based on original cost, and not the transfer price.

Adjustment to Opening Retained Earnings

When a transaction is recognized by a legal entity in one period and by the economic entity in another period, the straddling of the transaction across different periods requires consolidation adjustments to be passed to *opening retained earnings*. For example, Subsidiary Company sells inventory to Parent Company and makes a profit of $20,000 in 20x1. Parent Company resells 10% of the inventory to third parties in 20x1. Only 10% of the profit is earned by the group in 20x1. The opening retained earnings of Subsidiary Company in 20x2 includes "unrealized" profit of $18,000.

Opening retained earnings are disclosed as a separate line item in the consolidated statement of changes in equity. It then follows that the opening retained earnings in the current period's consolidated statement of changes in equity should agree exactly with the closing retained earnings as reported in the previous period's consolidated statement of changes in equity. In our example, the consolidated retained earnings at the end of 20x1 and the beginning of 20x2 should include the profit of $2,000 and not $20,000.

As discussed in Chapter 3, consolidation is a line-by-line summation of the items in the separate financial statements of the individual entities, which is then adjusted by consolidation entries. The sum of the opening retained earnings of the legal entities is not equal to the consolidated opening retained earnings because of timing differences in income recognition between the legal and economic entity. Hence, consolidation adjustments must be passed through again in the current year to ensure that the consolidated opening retained earnings is equal to the consolidated retained earnings at the previous year-end.[2] For example, unrealized profit from an intragroup transaction that is adjusted in one financial period is adjusted against opening retained earnings in the period immediately following it. The re-enactment continues for as long as the unrealized profit remains in retained earnings. If part of the inventory is resold to third parties, the amount in the re-enacted entry will be pro-rated accordingly. The application of this principle is explained in the illustrations in this chapter.

Tax Effects on Adjustments to Eliminate Profit or Loss in Carrying Amounts of Assets

Taxes on unrealized profits or losses have to be adjusted to align the *consolidated tax expense* with the *consolidated profit or loss*. When unrealized profits are eliminated from consolidated profit, but are included in the legal entity's taxable income, a *deferred tax asset* arises. IAS 12 describes the unrealized profit as a "deductible temporary difference" and requires the recognition of a deferred tax asset in the consolidated statement of financial position (IAS 12 Illustrative Examples B:11). Deferred tax asset is a type of prepaid tax. Although the profit is included in the taxable amount currently in the legal entity's tax returns, the tax on the unrealized profit is not an expense for group reporting. Hence, the tax expense relates to a future period and is deemed a deferred tax asset on the current period's statement of financial position. A consolidation adjustment is passed through to reclassify the tax expense to deferred tax asset in the consolidated statement of financial position. From a group's perspective, the tax expense is recognized only when the asset is resold to an external party or consumed or expensed off. The tax expense reported in the consolidated income statement should be aligned with the recognized profits for the economic entity. The consolidation adjustment in the year of transfer is as follows:

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Dr Deferred tax asset
  Cr Tax expense

In the year when the inventory is resold to third parties, the tax expense is realized in the consolidated profit and loss:

Dr Tax expense
  Cr Opening retained earnings

The above is a combination of two entries. First, the entry in the previous year is re-enacted as follows in the current year. The entry to tax expense in the previous year is taken to opening retained earnings in the current year.

Dr Deferred tax asset
  Cr Opening retained earnings

Next, an entry is shown to effect the crystallization of the deferred tax asset (a prepaid tax account) into an actual tax expense as the unrealized profit is earned.

Dr Tax expense
  Cr Deferred tax asset

The net effect of the two entries will give rise to the composite entry as shown above.

Conversely, when unrealized losses resulting from intragroup transactions are deducted for tax purposes by the legal entity, the reduction in the tax expense in the current period of the legal entity should not be recognized as such in the consolidated financial statements. The unrealized loss is deemed as a "taxable temporary difference" to the group (IAS 12 Illustrative Examples A:14), and a *deferred tax liability* should be recognized. The effect of this treatment is to increase the consolidated tax expense in the current period of the group to align with the higher consolidated profit arising from the adjustment of the unrealized loss.[3]

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The consolidation adjustment in the year of transfer is as follows:

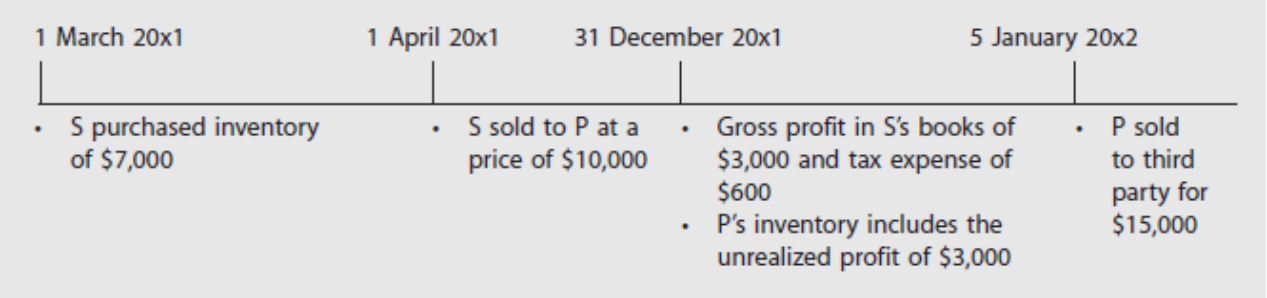
Dr Tax expense

Cr Deferred tax liability

The principles governing the elimination and adjustment of unrealized profit and loss are explained in the following illustrations. Illustration 5.1 shows the consolidation adjustments and the impact on the consolidated financial statements in a situation where a subsidiary sells inventory to its parent.

ILLUSTRATION 5.1 Upstream sale

During the financial year ended 31 December 20x1, the following sale was made to the parent company, P, by its 100% owned subsidiary, S. Assume a tax rate of 20%. For simplicity, assume that this was the only transaction that transpired over the two years for both companies. (In this example, we ignore the effect on non-controlling interests because S is a wholly owned subsidiary of P. The impact on non-controlling interests is considered in a subsequent section.)



Consolidation adjustments must be understood with reference to the original entries that are passed in the legal entities' separate financial statements. Hence, in the following consolidation adjustments, the legal entity, whose accounts are adjusted in the consolidation worksheet, is indicated in brackets. When the journal entry creates a new asset or liability for the group, the term "group" is indicated in brackets. The description in brackets is presented here to enhance understanding of the adjustment process but need not be presented otherwise.

Consolidation adjustments at 31 December 20x1

Dr Sales (S's I/S)	10,000	
Cr Cost of sales (S's I/S)		7,000
Cr Inventory (P's SFP)		3,000

Explanatory note:

This entry is passed to reduce current year profits and the overstatement of inventory because the gross profit of $3,000 is not yet realized from the group's perspective. The sales and cost of sales are entries in S's books, which are reversed out in the consolidation worksheet as they are not matched by offsetting entries in P. The inventory resides in P's statement of financial position; this entry removes the profit element from the book value of P's inventory.

Dr Deferred tax asset (Group SFP)	600	
Cr Tax expense (S's I/S)		600

Explanatory note:

As the profit is deemed as unearned in 20x1, the tax on the unrealized profit should not be recognized in 20x1; instead, the tax expense in S's books is considered as prepaid from the group's perspective, and is recognized as a deferred tax asset in accordance with IAS 12 *Income Taxes*. Tax expense should be aligned with income recognition.

Consolidation adjustments at 31 December 20x2

Dr Opening retained earnings (S's SFP)	3,000	
Cr Cost of sales (P's I/S)		3,000

Explanatory note:

This entry is passed to reduce the overstatement in the subsidiary's opening retained earnings. The entry shifts profits from the prior year's retained earnings to the current year's income when a sale to a third party is made. Note that S had recorded the profit as earned in its legal entities' financial statements in 20x1. Hence, the profit would be included in the opening retained earnings of S. A debit to opening retained earnings is necessary to remove the unrealized profit in S's retained earnings as at 1 January 20x2. A credit to cost of sales is necessary to reinstate the cost of sales recorded by P to the original purchase price of the inventory.

Let us consider the impact of a sale on tax expense in 20x2.

Dr Tax expense (Group's P/L)	600	
Cr Opening retained earnings (S's SFP)		600

Explanatory note:

Since the profit was realized in 20x2 through the external sale, tax expense should be recognized in the group income statement. As the tax expense was debited to S's income statement in 20x1, the opening retained earnings of S would have been understated by $600 from the group's perspective. The credit to opening retained earnings corrects the understatement while the debit to tax expense shows that the tax expense is now incurred for the group.

Alternatively, we can put through this pair of entries:

Dr Deferred tax asset (Group's SFP)	600	
Cr Opening retained earnings (S's SFP)		600
Dr Tax expense (Group's I/S)	600	
Cr Deferred tax asset (Group's SFP)		600

Let us consider a "what if" question to anchor the concepts further: What if the sale to an external party is made only in 20x3?

The debit entry to opening retained earnings is required to re-enact 20x1's entry, but the credit is now made to inventory as the profit is unrealized and the inventory of P is still overstated as at 31 December 20x2.

Dr Opening retained earnings (S's SFP)	3,000	
Cr Inventory (P's SFP)		3,000

If the external sale was not consummated in 20x2, the consolidation adjustment reinstates the prepaid tax and implicitly shifts the tax expense from the past period (as reflected in opening retained earnings) to the future period (as embodied in the deferred tax asset).

Dr Deferred tax asset (Group's SFP)	600	
Cr Opening retained earnings (S's SFP)		600

If we assume that this is the only transaction that transpired in the financial statements of P and S, the consolidation worksheets for the group for 20x1 and 20x2 are as follows:

Consolidation worksheet for 20x1

Consolidated Income Statement
For the Year Ended 31 December 20x1

	Combined P and S	Consolidation adjustment Dr	Consolidation adjustment Cr	Consolidated I/S
Sales	$10,000	$10,000		0
Cost of sales	(7,000)		$7,000	0
Gross profit	$ 3,000			0
Tax expense	(600)		600	0
Profit after tax	$ 2,400	$10,000	$7,600	0

Consolidated Statement of Financial Position
As at 31 December 20x1 (an extract)

	Combined P and S	Consolidation adjustment Dr	Consolidation adjustment Cr	Consolidated SFP
Cash	$ 2,400			$ 2,400
Inventory	10,000		$ 3,000	7,000
Deferred tax asset		$ 600		600
	$12,400	$ 600	$ 3,000	$10,000
Bank overdraft	$10,000			$10,000
Retained earnings	2,400	$10,000	$ 7,600	0
	$12,400	$10,000	$ 7,600	$10,000
Total adjustments		$10,600	$10,600	

The consolidated totals for 20x1 show that:

1. Inventory is carried at cost and not the marked-up price;
2. No profit is reported in the absence of any external sale; and
3. Tax expense is reclassified from an expense item in the income statement to an asset on the statement of financial position.

Consolidation worksheet for 20x2

Consolidated Income Statement For the Year Ended 31 December 20x2				
	Combined P and S	Consolidation adjustment		Consolidated I/S
		Dr	Cr	
Sales	$15,000			$15,000
Cost of sales	(10,000)		$3,000	(7,000)
Gross profit	$ 5,000			$ 8,000
Tax expense	(1,000)	$ 600		(1,600)
Profit after tax	$ 4,000			$ 6,400
Retained earnings, 1 January 20x2	2,400	3,000	600	0
Retained earnings, 31 December 20x2	$ 6,400	$3,600	$3,600	$ 6,400

Consolidated Statement of Financial Position As at 31 December 20x2 (an extract)				
	Combined P and S	Consolidation adjustment		Consolidated SFP
		Dr	Cr	
Cash	$6,400			$6,400
Retained earnings	$6,400	$3,600	$3,600	$6,400

The consolidated totals for 20x2 show that:

1. Reported sales amount is the external sales price, and the cost of sales is the original purchase price;
2. Tax expense is aligned with the reported profit; and
3. Opening retained earnings is zero. This position is consistent with the closing consolidated retained earnings from the previous year indicating a shift of profit from the previous year to the current year.

IMPACT ON NON-CONTROLLING INTERESTS ARISING FROM ADJUSTMENTS OF UNREALIZED PROFIT OR LOSS

Illustration 5.1 assumes a wholly owned subsidiary. In partially owned subsidiaries, the direction of the sale matters with respect to determining non-controlling interests' share of profits. Essentially, there are two types of sales, which are generally labelled as "downstream" or "upstream" sales.

Situation 1: "Downstream Sale"

When a parent sells to a subsidiary, that sale is commonly described as a "downstream sale," whereby the unrealized profit from the intercompany sale resides in the parent's books and the marked-up inventory remains on the subsidiary's statement of financial position. In such a situation, non-controlling interests' share of the profit of the subsidiary is not affected because the adjustment affects the parent's profits and not the subsidiary's.

Situation 2: "Upstream Sale"

If the sale is from a subsidiary to a parent, the profit that resides in the subsidiary's books is deemed as "unrealized" from the perspective of both parent and non-controlling interests. In this case, non-controlling shareholders assume the same perspective as the group. Since IFRS 10 paragraph B86(c) requires that "profits and losses resulting from intragroup transactions that are recognized in assets, such as inventory and fixed assets, are eliminated in full", it follows that the non-controlling interests' share of the unrealized profit or loss should also be adjusted from the carrying amount of the asset. This consistent treatment of unrealized profit or loss for both parent and non-controlling interests is an application of the entity theory of consolidation. This treatment also applies if one subsidiary sells to another subsidiary.

Illustration 5.2 explains the consolidation adjustments that have to be passed for upstream and downstream sales of inventory in a two-year setting.

ILLUSTRATION 5.2 Multi-period effects of "upstream" and "downstream" sales

P invested in 70% of the shares of S at its incorporation on 1 January 20x3. The table below provides information on intercompany transfers of inventory. Assume a tax rate of 20%. Net profit after tax of S was $800,000 for the year ended 31 December 20x3 and $900,000 for the year ended 31 December 20x4. No dividends were declared in 20x3 and 20x4. The transfers of inventory between P and S are shown below:

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	20x3	20x4
Sales of inventory from P to S	$ 60,000	
Original cost of inventory	(50,000)	
Gross profit	$ 10,000	
Percentage unsold to third parties at year-end	10%	4%
Sales of inventory from S to P	$200,000	
Original cost of inventory	(170,000)	
Gross profit	$ 30,000	
Percentage unsold to third parties at year-end	30%	0%

Required:

Prepare the consolidation adjustments for the year ended 31 December 20x3 and 31 December 20x4 to:

1. Eliminate intercompany sales;
2. Adjust unrealized profit in inventory;
3. Adjust the tax effects of the unrealized profit in inventory; and
4. Allocate share of current profit and opening retained earnings to non-controlling interests.

Consolidation adjustments at 31 December 20x3

CJE1: Elimination of intercompany sales and adjustment of unrealized profit from downstream sale

Dr Sales	60,000		
r Cost of sales		59,000	
r Inventory		1,000	(10% × $10,000)

Explanatory notes:

- Sales to S are fully eliminated as the consolidated financial statements should reflect only transactions with external parties.
- Cost of sales to be eliminated is derived as follows:

Cost of sales (as reported in P's I/S)	$50,000	
Cost of sales (as reported in S's I/S)	54,000	(90% of $60,000)
Combined cost of sales	104,000	
Cost of sales (from the group's perspective)	(45,000)	(90% of $50,000)
Amount to be eliminated	$59,000	

- The above working compares the combined cost of sales, which is the reported cost of sales from the legal entities' perspectives and the cost of sales from the group or economic entity's perspective. From the legal entity's perspective, the combined cost of sales is $104,000. However, from the economic entity's perspective, the internal cost of sales should not be recognized; we only want to include the cost of sales based on the original cost of the inventory for actual sales transacted with third parties. Since 90% of the inventory was sold to third parties, the consolidated cost of sales to be reported is 90% of the original cost of $50,000. Hence, we can analytically determine the cost of sales that should be reported. However, what is reported by the legal entities is $104,000. The difference is, thus, the consolidation adjustment.

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- Although consolidation seems complex, the adjustments are essentially a comparison of "what is reported" (the legal entity's perspective) against "what should be reported" (the economic entity's perspective) and adjusting for the difference. We have to keep in mind to always determine the reported numbers from an economic entity's perspective.
- The unrealized profit overstates the inventory balance at year-end. CJE1 adjusts the carrying amount of inventory to its original cost by removing the unrealized profit included in inventory. We can also analytically determine the consolidated balance of the inventory account. Remember that the reported inventory for the economic entity is the lower of original cost (not transfer price) and the net realizable value. However, for the legal entity, "cost" is the transfer price paid to another entity. The fact that the other entity is a group company is irrelevant to the stand-alone legal entity. Let us analytically determine the balance, which should be reported in the consolidated statement of financial position and the amount that is reported by the legal entity. Because the starting point of consolidation is the legal entity's financial statements, we will need to make an adjustment to remove the "profit" mark up from the unsold inventory. The following table shows the comparison as at 31 December 20x3.

Item	How it is calculated	Amount
Inventory as reported in the buying company's financial statements ("What is")	Percentage unsold × Transfer price	$6,000 (10% × $60,000)
Inventory, which should be reported in the consolidated financial statements ("What should be")	Percentage unsold × Original cost	$5,000 (10% × $50,000)
	Consolidation adjustment	$1,000

In CJE1, the cost of sales adjustment is determined as a residual. However, we should try to understand what this figure is made up of. An alternative way to present CJE1 is to pass two separate entries, CJE1(a) and CJE1(b). These two entries help to explain the adjustment to cost of sales.

- CJE1 is a composite entry of two sub-entries. For practical purposes, only one set of entries is required, but it is helpful to understand the nature of CJE1(a) and CJE1(b). If the composite entry CJE1 is used, we should pass through the sales and inventory entries first, and the cost of sales is a residual figure.

CJE1(a) eliminates the sales of P against the cost of sales of S for the proportion of inventory that was resold to third parties during 20x3.

CJE1(a): Elimination of realized sales from downstream sale		
Dr Sales (P)	54,000 (90% × $60,000)	
Cr Cost of sales (S)		54,000

Explanatory notes:

- Ninety percent of the sales by P to S is ultimately resold to third parties. Hence, this block of sales in P's financial statements is exactly offset by the cost of sales in S's financial statements. S would record the cost of sales at the price invoiced by P. Although there is no "bottom-line" effect on net profit, the group needs to put through this elimination entry to ensure that the disclosed sales and cost of sales figures in the consolidated financial statements are not overstated.
- There is no effect on the statement of financial position as there is no unrealized profit included in the inventory with respect to this block of sales.
- The tax effects with respect to the realized portion cancel out. As tax is levied on a legal entity basis, the profit in P's books will be subject to tax. However, the higher cost of sales in S's books will result in a higher tax deduction. As tax expense is a single line item in the consolidated financial statements, the offset between P's higher tax expense and S's lower tax expense ensures that the tax expense is not overstated.

CJE1(b) reverses the sales, cost of sales and profit in inventory for the proportion of inventory that remained unsold as at 31 December 20x3.

CJE1(b): Reversal of unrealized sales and removal of profit from inventory		
Dr Sales (P)	6,000 (10% × $60,000)	
Cr Cost of sales (P)		5,000 (10% × $50,000)
Cr Inventory (S)		1,000

Explanatory notes:

- A profit element of $1,000 from an intragroup transaction is included in the carrying amount of the inventory. The carrying amount of the inventory in the consolidated statement of financial position has to be adjusted to the lower of cost and net realizable value. Cost is the original purchase price as transacted with third parties.
- Hence, an adjustment is required to remove the unrealized profit from S's inventory in the statement of financial position. The adjustment essentially brings the carrying amount of inventory to the original cost, as recorded in P's books, before the transfer was made.

- This adjustment effectively reverses the sales and cost of sales of P on the unrealized portion, and puts the consolidated financial statements in a position as if the transfer never happened.

Hence, the rationale for CJE1(a) and CJE1(b) is different. CJE1(a) eliminates the sales of P against the cost of sales of S to ensure the reported figures in the consolidated income statement are not overstated. CJE1(b), on the other hand, reverses out the transaction altogether as the profit is not yet earned from a group's perspective. If we combine CJE1(a) and CJE1(b), we arrive at CJE1. Practically, there is no need to pass CJE1(a) and CJE1(b) separately, but it is meaningful to understand the reasons for adjusting cost of sales. Thus, the consolidation adjustment to cost of sales of $59,000 in CJE1 comprises two elements. We have explained the two elements in CJE1(a) and CJE1(b) and summarized the rationale and calculations in Table 5.2.

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TABLE 5.2 Rationale for the cost of sales adjustment in consolidation

Rationale	How it is calculated	Amount
Cost of sales of the resold portion of the buying company is eliminated against sales of the selling company because it is an intragroup transaction. The elimination is also required because there is double counting in cost of sales. The original cost of sales is already reported in the selling company's income statement.	Cost of sales eliminated = Percentage resold to third parties × Transfer price	$54,000 (90% × $60,000)
Cost of sales of the unsold portion of inventory is reversed out because the inventory has not been sold and still remains within the economic entity.	Cost of sales reversed = Percentage unsold × Original cost	$ 5,000 (10% × $50,000)
	Consolidation adjustment	$59,000

The next entry, CJE2, shows the consolidation adjustment for the tax effects on CJE1.

CJE2: Adjustment for the tax effects on unrealized profit in inventory from downstream sale

Dr Deferred tax asset	200	
Cr Tax expense		200 (20% × $1,000)

Explanatory notes:

- Since the unrealized profit is deemed as "unearned" from the group's perspective, no tax expense should be recognized on the profit in the consolidated income statement.
- Tax expense is overstated and reclassified to the statement of financial position as a deferred tax asset as it relates to a future period.

Illustration 5.2 also presents a situation where S sold inventory to P. When a subsidiary sells an asset to the parent, the transfer is referred to as an upstream sale. We need to eliminate the sales and cost of sales in the same manner as we eliminated the sales and cost of sales of the downstream sale in CJE1. The following entries CJE3 and CJE4 show the profit and tax adjustments, respectively.

CJE3: Elimination of intercompany sales and adjustment of unrealized profit from upstream sale

Dr Sales	200,000	
Cr Cost of sales		191,000

Cr Inventory		9,000	(30% × $30,000)

Explanatory notes:

- Sales to P of $200,000 have to be eliminated in full as this is an intragroup transaction.
- Inventory is overstated by $9,000 as it includes unrealized profit. We have to restate the inventory to the original cost as transacted with third parties. We can analytically determine the inventory balance in the consolidated statement of financial position and compare it with what is reported in the legal entity's financial statements, as follows:

Item	How it is calculated	Amount
Inventory as reported in the buying company's financial statements ("What is")	Percentage unsold × Transfer price	$60,000 (30% × $200,000)
Inventory, which should be reported in the consolidated financial statements ("What should be")	Percentage unsold × Original cost	$51,000 (30% × $170,000)
	Consolidation adjustment	$ 9,000

- Cost of sales is the residual figure in this elimination entry. However, we can analyze the residual in a few ways. First, we can compare the combined cost of sales in the legal entities' financial statements with the amount that should be reported by the economic entity.

Cost of sales (as reported in S's I/S)	$170,000	
Cost of sales (as reported in P's I/S)	140,000	(70% of $200,000)
Combined cost of sales	$310,000	
Cost of sales (from the group's perspective)	(119,000)	(70% of $170,000)
Amount to be eliminated	$191,000	

Alternatively, we can analyze the total consolidation adjustment to two sub-parts: the cost of sales of the resold portion that should be eliminated against intercompany sales and the amount of the cost of sales of the unsold portion that should be reversed as there is no external transaction. The following table shows how the two sub-parts add up to give the total consolidation adjustment. As explained earlier, this analysis is to help us understand the nature of the consolidation adjustment on cost of sales. However, practically, the cost of sales adjustment may be determined as a residual after the sales and inventory adjustments are made.

Rationale	How it is calculated	Amount
Cost of sales of the resold portion	Cost of sales eliminated = Percentage resold to third parties × Transfer price	$140,000 (70% × $200,000)
Cost of sales of the unsold portion of inventory	Cost of sales reversed = Percentage unsold × Original cost	$ 51,000 (30% × $170,000)
	Consolidation adjustment	$191,000

CJE4: Adjustment for the tax effects on unrealized profit in inventory from upstream sale

Dr Deferred tax asset	1,800	
Cr Tax expense		1,800 (20% × $9,000)

Explanatory note:

As with CJE2, the tax expense on the unrealized profit on the upstream sale has to be reclassified to deferred tax asset to align the recognition of tax expense with the underlying profit.

In this illustration, P owns 70% of S. Hence, the consolidation process has to attribute 30% of the net profit of S to non-controlling interests. The next entry CJE5 reflects the allocation of profit to non-controlling interests. Without this entry, the equity "bucket" belonging to the parent's shareholders will be overstated and the equity "bucket" of non-controlling interests will be understated.

CJE5: Allocation of current profit after tax to non-controlling interests

Dr Income to non-controlling interests (I/S)	237,840	
Cr Non-controlling interests (SFP)		237,840

Net profit after tax of S for 20x3	$800,000
Less unrealized profit from upstream sale (CJE3)	(9,000)
Add tax expense on unrealized profit from upstream sale (CJE4)	1,800
Adjusted net profit after tax of S for 20x3	$792,800
Non-controlling interests' share of profit after tax for 20x3 at 30%	$237,840

Explanatory notes:

- Non-controlling interests' share of profit after tax is adjusted for the unrealized profit on the upstream sale and its tax effect.
- Why is CJE5 required? Why do we not simply pro-rate CJE3 and CJE4 between non-controlling and parent's share of the unrealized profit and related tax expense? CJE5 is required because of the structure of the consolidated income statement. Consolidated profit after tax relates to the entity as a whole and is presented before the allocation to non-controlling interests. Hence, the total amount of adjustments must be effected to the consolidated profit after tax. Allocation to non-controlling interests, shown as a separate line item after profit after tax, incorporates non-controlling interests' share of the adjustments.

The next set of entries show the consolidation adjustments that are required in 20x4. Because the transfers of inventory were made in 20x3 with a portion resold in 20x4, a number of these entries would adjust opening retained earnings to remove unrealized profit as at 1 January 20x4.

Consolidation adjustments at 31 December 20x4

CJE1: Adjustment of unrealized profit from downstream sale in retained earnings as at 1 January 20x4

Dr Opening retained earnings	1,000 (10% × $10,000)	
Cr Cost of sales		600 (6% × $10,000)
Cr Inventory		400 (4% × $10,000)

Explanatory notes:

- P's opening retained earnings are overstated by the unrealized profit as at January 20x4. Recall that 10% of the gross profit of P of $10,000 was "unrealized" as 10% of the inventory remained within the group at 31 December 20x3. This entry has to be passed through each year for as long as the inventory is unsold.
- In this situation, the inventory is only partially sold by 31 December 20x4. Four percent remained unsold while 6% was sold to third parties.

Item	How it is calculated	Amount
Cost of sales as reported in the buying company's financial statements ("What is")	Percentage resold × Transfer price	$3,600 (6% × $60,000)
Cost of sales, which should be reported in the consolidated financial statements ("What should be")	Percentage resold × Original cost	$3,000 (6% × $50,000)
	Consolidation adjustment	$600

The profit on the sold portion is credited to cost of sales to adjust the cost of sales to the original cost. Simply put, we can say that 6% of the gross profit of P is "realized" or earned because the inventory has been resold to third parties. Hence, 6% of the original gross profit of $10,000 is credited to the income statement. On another angle, we can compare the cost of sales at two levels (legal and economic entity) and make the appropriate adjustment.

- The unrealized profit at the end of the year is credited to inventory to adjust the inventory to its original cost. Four percent of the gross profit of P should still not be recognized because the inventory remains within the group. Hence, an adjustment of $400 or 4% of $10,000 is effected to inventory.

CJE2: Adjustment of tax on unrealized profit from downstream sale in retained earnings as at 1 January 20x4

Dr Tax expense	120	
Dr Deferred tax asset	80	
Cr Opening retained earnings		200

Explanatory note:

This entry reflects the tax effects of CJE1.

CJE3: Allocation of post-acquisition retained earnings as at 1 January 20x4

Dr Opening retained earnings	240,000	
Cr Non-controlling interests		240,000

Explanatory note:

Since 20x3 is the first year of incorporation, the share of profit is 30% of $800,000 (assuming no dividends are paid) is allocated to non-controlling interests. We use the unadjusted profit after tax as a base to compute non-controlling interests' share of post-acquisition retained earnings. The adjustments for unrealized profit and tax on unrealized profit are shown separately below in CJE4 and CJE5. The combined effect of CJE3, CJE4, and CJE5 results in non-controlling interests' share of adjusted opening retained earnings.

CJE4: Adjustment of unrealized profit from upstream sale in retained earnings as at 1 January 20x4

Dr Opening retained earnings	6,300 (70% × 30% × $30,000)	
Dr Non-controlling interests	2,700 (30% × 30% × $30,000)	
Cr Cost of sales		9,000 (30% × $30,000)

Explanatory notes:

- Opening retained earnings are adjusted with the parent's share of the unrealized profit adjustment. Note that retained earnings as reported in the consolidated financial statements belong to the parent's shareholders. Hence, non-controlling interests' share is shown separately.
- Since non-controlling interests adopt the same perspective as the parent with respect to the elimination of profit on upstream sale, non-controlling interests are debited for their share of unrealized profit in inventory at the start of the year. This entry re-enacts the adjustment to non-controlling interests' share of profit in the previous year. Recall that non-controlling interest is a composite equity item and includes non-controlling interests' share of retained earnings.
- The credit entry reflects the event(s) that transpires in the current year with respect to the inventory that was transferred in the previous period. The inventory is sold; hence, a credit entry to cost of sales is necessary to adjust P's cost of sales to the original cost. We can now recognize the unrealized profit as at 31 December 20x3 in full in 20x4. Recall that the unrealized profit is $9,000, or 30% of $30,000. If we analyze the cost of sales at the two levels, we will arrive at a differential that is $9,000. The following table explains the differential.

Item	How it is calculated	Amount
Cost of sales as reported in the buying company's financial statements ("What is")	Percentage resold × Transfer price	$60,000 (30% × $200,000)
Cost of sales, which should be reported in the consolidated financial statements ("What should be")	Percentage resold × Original cost	$51,000 (30% × $170,000)
	Consolidation adjustment	$9,000

CJE5: Adjustment of tax on unrealized profit from upstream sale as at 1 January 20x4

Dr Tax expense	1,800	
Cr Opening retained earnings		1,260
Cr Non-controlling interests		540

Explanatory note:

CJE5 shows the tax effects of CJE4.

CJE6: Allocation of current profit after tax to non-controlling interests

Dr Income to non-controlling interests (I/S)	272,160	
Cr Non-controlling interests (SFP)		272,160
Net profit after tax of S for 20x4		$900,000
Add realized profit from upstream sale (CJE4)		9,000
Less tax expense on realized profit from upstream sale (CJE5)		(1,800)
Adjusted net profit after tax of S for 20x4		$907,200
Non-controlling interests' share of profit after tax for 20x4		$272,160

Explanatory notes:

- CJE6 shows the delayed recognition of previously unrecognized profit on the intercompany upstream transfer. Since the inventory is resold to third parties, the profit is earned and the realized profit and tax effects are added to the net profit after tax of S.
- You should note that only realized profit and tax effects arising from the current sale of inventory transferred from group companies in *previous* periods are added back. Conversely, only the unrealized profit and tax effects relating to unsold inventory transferred from group companies in the *current* period are deducted from the current year profit after tax.

SPECIAL CONSIDERATIONS FOR INTERCOMPANY TRANSFERS OF FIXED ASSETS

When fixed assets are transferred at a marked-up price between group companies, the unrealized profit (or loss) must be eliminated from the carrying amount of the fixed assets. From the group's point of view, the fixed assets remain within the group and it must account for the fixed assets as if the transfer did not take place at all. The following adjustments are, therefore, necessary:

1. The carrying amount of the fixed asset must be reinstated to the original cost and accumulated depreciation of the selling company as of the date of the transfer.
2. The profit or loss on the intercompany sale of the fixed asset must be adjusted out of the consolidated income statement, if the sale occurred in the current period. If the sale occurred in a prior period, the adjustment is made against consolidated retained earnings with consequential impact on non-controlling interests, provided the transfer is an upstream sale.
3. Depreciation subsequent to the transfer must be determined on the basis of the original historical cost of the asset and the assets estimated useful life (incorporating any change in estimates subsequent to purchase). If the estimated useful life has changed, the group depreciation should be the net book value based on original cost divided by the estimated useful life. Essentially, the "new" depreciation that is expensed to the legal entity's financial statements is calculated on the basis of the transfer price. The difference between the legal entity depreciation and the group depreciation for the current period must be adjusted to the consolidated income statement. The cumulative effects of the adjustments to depreciation relating to prior periods must be adjusted to consolidated retained earnings.
4. The profit or loss on the intercompany sale of the fixed asset is "realized" through the process of depreciation. Unlike inventory, which is held for sale, fixed assets are held for the production of goods and services. Hence, the cost of the fixed asset is consumed through usage over the remaining useful life. From a group's perspective, unrealized profit or loss arising from the transfer of the fixed asset artificially inflates (or reduces)

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the cost of the asset. The excess is "realized" through a series of higher depreciation charges in periods after the transfer. Over the life of the asset, the aggregate of the additional depreciation equals the "profit" on the sale.

5. How is the profit realized through depreciation? We explained earlier that the profit on sale of fixed asset is "realized" through the depreciation process. Essentially, the profit earned by the selling company will be offset by the higher depreciation incurred by the buying company. Since both companies are in the same "family", it is simply a case that one member has higher earnings while another has lower earnings. However, the reduction in earnings of the buying company arises over current and future periods and thus, a consolidation adjustment has to be passed through to remove the unrealized profit and to correct the excess depreciation. From a group's perspective, the profit on sale is only a timing difference that will unwound through excess depreciation. We show it by way of an analytical working:

Profit on sale of fixed assets = Proceeds (TP) – Net book value (NBV) of fixed assets

In a transfer of fixed assets between group companies, proceeds will be the transfer price (TP). We compare the legal entity depreciation ("what is") and the group depreciation ("what should be").

Legal entity depreciation = TP/Remaining Useful Life

Group depreciation = NBV/Remaining Useful Life

(In calculating the group depreciation, the most up-to-date useful life estimates should be used. Hence, we do not use the original useful life, which may be redundant at the point of transfer. The numerator for group depreciation is the net book value at the point of transfer. The net book value is the remaining undepreciated cost.)

$$\begin{aligned}\text{Excess depreciation} &= \text{Legal entity depreciation} - \text{Group depreciation}\\ &= \text{TP/Remaining Useful Life} - \text{NBV/Remaining Useful Life}\\ &= \text{(TP} - \text{NBV)/Remaining Useful Life}\\ &= \text{Profit on sale of fixed assets/Remaining Useful Life}\end{aligned}$$

From the above analysis, we arrive at the conclusion that:

Profit on sale of fixed assets = Excess depreciation × Remaining useful life

6. What if the transferred fixed asset is disposed or rendered obsolete before the end of the useful life? The "realization" is then accelerated and crystallized at the event of the sale or impairment. Our analytical equation will then be revised. If the transferred fixed asset is resold to third parties, the equality is as follows:

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Profit on sale of fixed assets = (Excess depreciation × Expired periods) + Excess profit on resale

$$\begin{aligned}\text{Excess profit on resale} &= \text{Legal entity profit on sale} - \text{Group profit on sale}\\ &= \text{Proceeds} - [\text{TP/Remaining useful life} \times \text{Unexpired life}]) -\\ &\quad (\text{Proceeds} - \text{NBV/Remaining useful life} \times \text{Unexpired life})\\ &= \text{(TP} - \text{NBV)/Remaining useful life} \times \text{Unexpired life}\end{aligned}$$

If the transferred fixed asset is impaired, the carrying amount of the fixed asset will be written down to the recoverable amount. A very simple approach to figuring out what to adjust is to ask this question: What is the difference between the carrying amount of the asset of the legal entity and the group? If both the legal and the

group entity have to write down the carrying amount to the recoverable amount, there will be no difference in the reported balances in the legal entity and group financial statements. Hence, the "unrealized" profit is unwound totally at this point when the carrying amount at the two levels converges. In the later section on the transfer of assets at a loss, we discuss the interactions among original cost, transfer price, and recoverable amount.

7. Tax effects should be adjusted in line with the adjustments on unrealized profit or loss and the subsequent corrections of depreciation. Tax effects on unrealized profit are accounted as deferred tax asset in the consolidated statement of financial position. The tax effects on the correction of the excess depreciation will set off the deferred tax asset over the remaining useful life.
8. The principles and processes that relate to the adjustment of profit on transfer of fixed assets between group companies also apply to other long-lived assets such as intangible assets.

If fixed assets are carried at revalued amounts, other comprehensive income arising from the revaluation must be determined on the basis of the original cost of the fixed assets as transacted with the external supplier. Consolidation adjustments are required to measure other comprehensive income from the economic entity's perspective.

Impact on Non-controlling Interests when an Unrealized Profit Arises from an Intragroup Transfer of Fixed Assets

If the transfer is a downstream sale, the correction of the profit (or loss) and the correction of excess (or shortfall) in depreciation have no impact on the allocation of current profit after tax to non-controlling interests. The elimination of the unrealized profit from the carrying amount of the fixed assets will apply only to the parent. Illustration 5.3 explains the consolidation adjustments relating to such a transfer.

If the transfer is an upstream sale, the correction of the profit (or loss) and the correction of excess (or shortfall) in depreciation will be adjusted to profit to determine the non-controlling interests' share. Adjustments to unrealized profit and cumulative depreciation of past periods from upstream transfers are allocated to non-controlling interests as well.

Non-controlling interests' share of profit is adjusted for any excess or shortfall in depreciation arising from an upstream sale of fixed assets. As explained above, unrealized profit on the sale of fixed assets is realized through the process of depreciation. Hence, the non-controlling interests' share of unrealized profit from upstream sales is unwound through the subsequent depreciation on the unrealized profit. It does not matter that the depreciation expense is recorded in the parent's books. The two transactions (unrealized profit in the subsidiary's books and the depreciation expense in the parent's books) are linked and would thus have an impact on non-controlling interests. Tax effects on the profit and depreciation adjustments are also adjusted to non-controlling interests.

ILLUSTRATION 5.3 "Downstream" transfer of fixed assets

On 1 January 20x2, Parent (P) sold equipment to Subsidiary (S) for $360,000. The original cost of the equipment was $400,000, which was depreciable over ten years from 1 January 20x0. S depreciated the equipment over the remaining useful life of eight years from 1 January 20x2. Assume a tax rate of 20%.

$$\begin{aligned}\text{Profit on sale recorded by P} &= \text{Sales price} - \text{Net book value}\\ &= \$360{,}000 - (\$400{,}000 - \$80{,}000)\\ &= \$360{,}000 - \$320{,}000\\ &= \$40{,}000\end{aligned}$$

Entry in P's books

Dr Cash	360,000	
Dr Accumulated depreciation	80,000	
Cr Equipment, cost		400,000
Cr Profit on sale		40,000

Profit on sale of fixed assets to S

Entry in S's books

Dr Equipment	360,000	
Cr Cash		360,000

Purchase of fixed assets from P

Dr Depreciation expense	45,000	
Cr Accumulated depreciation		45,000

Depreciation expense for 20x2

As at 31 December 20x2

	Status quo	With sale	Amount to be restored/ adjusted	
Cost of asset	$400,000	$360,000	$40,000	
Accumulated depreciation	120,000	45,000	75,000	80,000
Current depreciation	40,000	45,000	5,000	
Profit on sale	–	40,000	40,000	
Tax on profit	–	8,000	8,000	

The consolidation adjustments are shown below. To enhance clarity, each reversing or adjusting entry shows, in brackets, the entity whose balances have to be adjusted in the worksheet:

Consolidation adjustments at 31 December 20x2

CJE1: Adjustment of unrealized profit

Dr Equipment (S)	40,000	
Dr Profit on sale (P)	40,000	
Cr Accumulated depreciation (S)		80,000

This entry reinstates the equipment to original cost and the accumulated depreciation as at the date of transfer and reverses the profit. (The debit adjustment of $40,000 increases the cost of fixed assets for the group to the original cost of $400,000. The credit entry reinstates accumulated depreciation to $80,000, which is the balance in P's books prior to the transfer. Accumulated depreciation of $80,000 is made up of two years of depreciation ($400,000/10 × 2) from 1 January 20x0 to 1 January 20x2.)

CJE2: Reverse tax on profit on sale

Dr Deferred tax asset (Group's SFP)	8,000	
Cr Tax expense (P)		8,000

CJE3: Correct the over-depreciation on unrealized profit included in equipment

Dr Accumulated depreciation (S)	5,000	
Cr Depreciation (S)		5,000

Depreciation recorded by S in 20x2 ($360,000/8)	$45,000
Original depreciation had P not sold to S ($400,000/10)	40,000
Excess depreciation corrected for group reporting	$ 5,000

Alternatively: Excess depreciation = Unrealized profit/Remaining useful life
= $40,000/8
= $5,000

CJE4: Increase in tax arising from correction of over-depreciation

Dr Tax expense (S)	1,000	
Cr Deferred tax asset (Group's SFP)		1,000

Consolidation adjustments at 31 December 20x3

CJE1: Adjustment of unrealized profit in the prior year

Dr Equipment (S)	40,000	
Dr Opening retained earnings (P)	40,000	
Cr Accumulated depreciation (S)		80,000

Non-controlling interests are not affected because the transfer is a downstream sale

CJE2: Reverse tax on profit on sale arising in the prior year

Dr Deferred tax asset (Group's SFP)	8,000	
Cr Opening retained earnings (P)		8,000

CJE3: Reverse excess depreciation for 20x2

Dr Accumulated depreciation (S)	5,000	
Cr Opening retained earnings (S)		5,000

CJE4: Increase in tax from reversal of excess depreciation for 20x2

Dr Opening retained earnings (S)	1,000	
Cr Deferred tax asset (Group's SFP)		1,000

CJE5: Correct the current over-depreciation on unrealized profit included

in equipment

Dr Accumulated depreciation (S)	5,000	
Cr Depreciation (S)		5,000

Depreciation recorded by S in 20x3 ($360,000/8)	$45,000
Original depreciation had P not sold to S ($400,000/10)	40,000
Excess depreciation corrected for group reporting	$ 5,000

CJE6: Increase in tax arising from correction of over-depreciation

Dr Tax expense (S)	1,000	
Cr Deferred tax asset (Group's SFP)		1,000

The correction of the excess depreciation continues on a yearly basis until the asset is fully depreciated. Since fixed assets are not held for sale, the unrealized profit on the sale of fixed assets is deemed "realized" when the asset is fully depreciated. When the asset is fully depreciated, the following consolidation adjustments will offset each other at, and after, the end of the eight years from the date of transfer:

CJE1: Reinstate to original cost and accumulated depreciation and reverse profit

Dr Equipment (S)	40,000	
Dr Opening retained earnings (P)	40,000	
Cr Accumulated depreciation (S)		80,000

CJE2: Correction of past excess depreciation

Dr Accumulated depreciation (S)	40,000	
Cr Opening retained earnings (S)		40,000

$5,000 × 8 = $40,000

The tax effects will also offset each other fully, assuming a tax rate of 20%.

CJE3: Tax effects on unrealized profit on sale of fixed assets

Dr Deferred tax asset	8,000	
Cr Opening retained earnings (P)		8,000

CJE4: Tax effects on correction of excess depreciation

Dr Opening retained earnings (S)	8,000	
Cr Deferred tax asset		8,000

What if the transfer of the fixed asset is from a partially owned subsidiary to its parent? Non-controlling interests will have to bear a proportion of the adjustments. Consider the impact on the consolidation journal entries in an extension

to Illustration 5.3 where the transfer is featured as an upstream sale.

ILLUSTRATION 5.3 EXTENSION "Upstream" transfer of fixed assets

On 1 January 20x2, Subsidiary (S) sold equipment to Parent (P) for $360,000. The original cost of the equipment was $400,000, which was depreciable over ten years from 1 January 20x0. P depreciated the equipment over the remaining useful life of eight years from 1 January 20x2. Net profit after tax of S for the year ended 31 December 20x2 and 31 December 20x3 was $500,000 and $800,000 respectively. P owned 90% of S. Assume a tax rate of 20%.

Consolidation adjustments at 31 December 20x2

CJE1: Adjustment of unrealized profit		
Dr Equipment (P)	40,000	
Dr Profit on sale (S)	40,000	
Cr Accumulated depreciation (P)		80,000
CJE2: Reverse tax on profit on sale		
Dr Deferred tax asset (Group's SFP)	8,000	
Cr Tax expense (S)		8,000

CJE3: Correct the current over-depreciation on unrealized profit included in equipment		
Dr Accumulated depreciation (P)	5,000	
Cr Depreciation (P)		5,000
Depreciation recorded by P in 20x2 ($360,000/8)		$45,000
Original depreciation had S not sold to P ($400,000/10)		40,000
Excess depreciation corrected for group reporting		$ 5,000
CJE4: Increase in tax arising from correction of over-depreciation		
Dr Tax expense (P)	1,000	
Cr Deferred tax asset (Group's SFP)		1,000

CJE5: Allocation of current year income to non-controlling interests			
Dr Income to non-controlling interests	47,200		
Cr Non-controlling interests		47,200	
Net profit after tax of S for year ended 31 December 20x2		$500,000	
Less unrealized profit on sale	$(40,000)		
Add tax expense on unrealized profit on sale	8,000	(32,000)	(Note 1)
Add realization through depreciation	5,000		(Note 1)
Less tax expense on depreciation	(1,000)	4,000	(Note 1)
Adjusted net profit after tax of S for 20x2		$472,000	
Non-controlling interests' share at 10%		$ 47,200	

Note 1: Although the excess depreciation is recorded in P's books, it is a realization of the profit originating from S's books as depreciation depicts the consumption of the benefits of the fixed asset. The correction of the excess depreciation affects the non-controlling interests' share of profit. Depreciation (net of tax) will "unwind" the original profit on sale (net of tax) until the end of the remaining useful life of eight years is reached.

Consolidation adjustments at 31 December 20x3

CJE1: Adjustment of unrealized profit in the prior year				
Dr Equipment (P)	40,000			
Dr Opening retained earnings (S)	36,000	(90% × $40,000)		
Dr Non-controlling interests	4,000	(10% × $40,000)	(Note 2)	
Cr Accumulated depreciation (P)				80,000

Note 2: Since this is an upstream sale, non-controlling interests share a proportion of the adjustment to unrealized profit arising from the prior period. Adjustment to opening retained earnings reflects only the parent's share.

CJE2: Reverse tax on profit on sale for prior year			
Dr Deferred tax asset (Group's SFP)	8,000		
Cr Opening retained earnings (S)		7,200	(20% × $36,000)
Cr Non-controlling interests		800	(20% × $4,000)

CJE3: Reverse excess depreciation for prior year				
Dr Accumulated depreciation (P)	5,000			
Cr Opening retained earnings (P)		4,500	(90% × $5,000)	
Cr Non-controlling interests		500	(10% × $5,000)	(Note 3)

Note 3: Non-controlling interests bear a share of the reversal entry relating to excess depreciation, since depreciation is a systematic amortization of the unrealized profit.

CJE4: Increase in tax from reversal of excess

depreciation for 20x2 (prior year)

Dr Opening retained earnings (P)	900 (20% × $4,500)	
Dr Non-controlling interests	100 (20% × $500)	
Cr Deferred tax asset (Group's SFP)		1,000 (20% × $5,000)

CJE5: Correct the current over-depreciation on unrealized profit included in equipment

Dr Accumulated depreciation (P)	5,000	
Cr Depreciation (P)		5,000

Depreciation recorded by P in 20x3 ($360,000/8)	$45,000
Original depreciation had S not sold to P ($400,000/10)	40,000
Excess depreciation corrected for group reporting	$ 5,000

CJE6: Increase in tax arising from correction of over-depreciation

Dr Tax expense (P)	1,000	
Cr Deferred tax asset (Group's SFP)		1,000

CJE7: Allocation of current year income to non-controlling interests

Dr Income to non-controlling interests (Group's I/S)	80,400	
Cr Non-controlling interests (Group's SFP)		80,400

Net profit after tax of S for year ended 31 December 20x3		$800,000
Add realization through depreciation	$5,000	
Less tax expense on depreciation	(1,000)	4,000
Adjusted net profit after tax of S for 20x3		$804,000
Non-controlling interests' share at 10%		$ 80,400

Subsequently, when the asset is fully depreciated, the following consolidation adjustments will offset each other at and after the end of the eight years from the date of sale:

CJE1: Adjustment of unrealized profit in a prior year

Dr Equipment (P)	40,000	
Dr Opening retained earnings (S)	36,000	
Dr Non-controlling interests	4,000	
Cr Accumulated depreciation (P)		80,000

CJE2: Correction of past excess depreciation

Dr Accumulated depreciation (P)	40,000	
Cr Opening retained earnings (P)		36,000

Cr Non-controlling interests		4,000

$5,000 × 8 = $40,000

Assuming a tax rate of 20%, the offsetting tax effects are as follows:

CJE4: Tax effects on unrealized profit on sale of fixed assets		
Dr Deferred tax asset	8,000	
Cr Opening retained earnings (S)		7,200
Cr Non-controlling interests		800
CJE5: Tax effects on correction of excess depreciation		
Dr Opening retained earnings (P)	7,200	
Dr Non-controlling interests	800	
Cr Deferred tax asset		8,000

In Chapter 4, we have explained the process to account for non-controlling interests and the analytical check that we can perform to test the balance of non-controlling interests. We expand the analytical check of non-controlling interests to include the effects of adjustments for upstream transfers. Figure 5.2 shows the three components of non-controlling interests.

FIGURE 5.2 Components of non-controlling interests

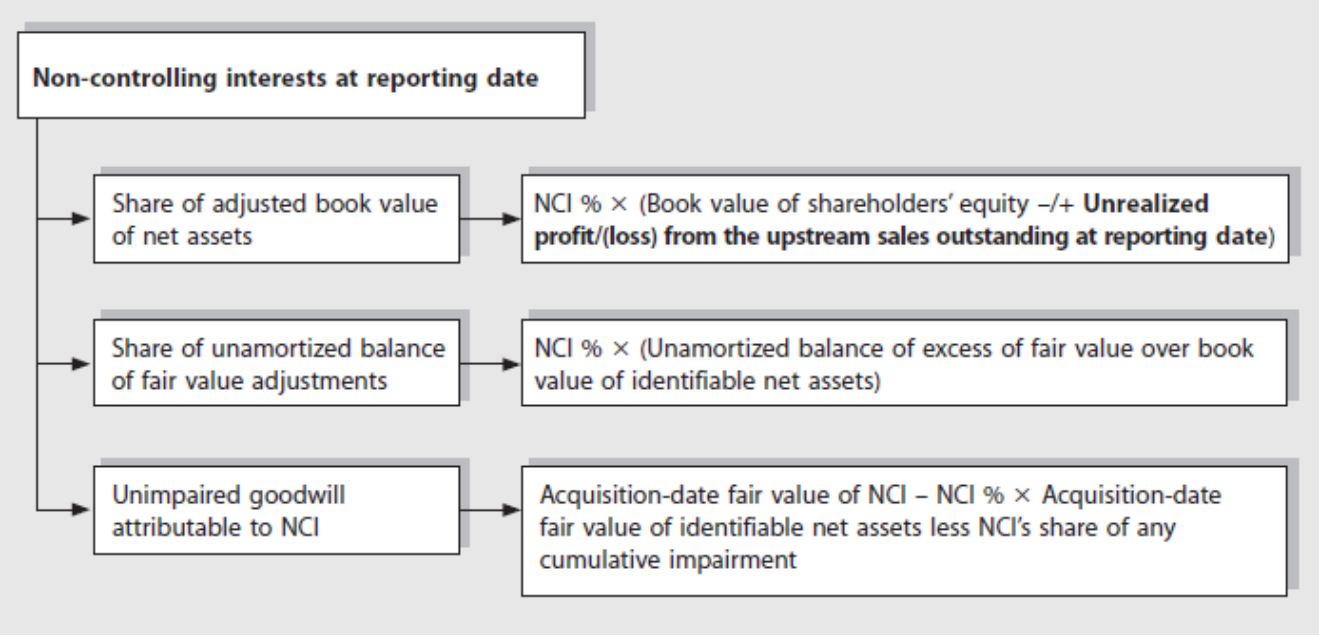


Table 5.3 illustrates the movement in non-controlling interests' balance.

TABLE 5.3 Analytical check on non-controlling interests' balance

Event	Components		
	Book value of identifiable net assets	**Fair value less book value of identifiable net assets**	**Goodwill**
Date of acquisition (re-enacted each year)	Share of book value of net assets or equity of subsidiary as at acquisition date	Share of fair value adjustments of identifiable net assets as at date of acquisition	Non-controlling interests' goodwill
From acquisition date to beginning of current period (re-enacted each year)	+/– Share of post-acquisition profit / (loss) of subsidiary –/+ Share of after-tax adjustments for unrealized profit/(loss) included in beginning retained earnings +/– Share of other changes in equity	–/+ Past cumulative amortization of fair value adjustments of identifiable net assets	– Past cumulative impairment of goodwill
Current period	+/– Share of profit/(loss) of subsidiary after tax –/+ Share of after-tax adjustments of unrealized profit/(loss) from upstream sales +/– Share of after-tax adjustments of realized profit/(loss) in current period in respect of previous period's upstream sales – Share of dividends +/– Share of other changes in equity	–/+ Current amortization of fair value adjustments of identifiable net assets	– Current impairment of goodwill
Non-controlling interests' balance at year-end	= Share of book value of net assets or equity of subsidiary at year-end **–/+ share of unrealized profit/(loss) at year-end**	= Share of unamortized balance of fair value adjustments of identifiable net assets at year-end	= Non-controlling interests' unimpaired goodwill

Analytical Check on Consolidated Retained Earnings

Thus far, we have considered an analytical check on the balance of non-controlling interests, which allows us to analyze the balance independently of the consolidation adjustments. These analytical checks are useful as they allow an expedient way to arrive at the same end result in an independent manner. That said, some common errors may affect the accuracy of both the consolidation adjustments and the independent analysis. For example, if the wrong ownership interest is applied in the calculations, both the process of the consolidation adjustments and the independent check of the end result will be in error.

We now want to consider another critical account balance, the consolidated retained earnings. The balance in retained earnings is the net result of transactions involving income and expenses. The retained earnings balance, at the start and end of year, would be subject to many consolidation adjusting entries. The likelihood of error is high and it is worthwhile to analytically determine the balance to substantiate the end result from the consolidation process.

As with the non-controlling interests balance, we can arrive at consolidated retained earnings in one of two ways. Through the consolidation process, we can list out all the consolidation adjustments that affect retained earnings. If

the focus is on the ending retained earnings balance, we will start with the summation of the parent's retained earnings and the subsidiaries' retained earnings, and list out all consolidation adjustments that affect opening retained earnings and current year income statement items. The listings approach simulates the process involved in compiling a consolidation worksheet to arrive at consolidated totals.

The analytical approach to determine the consolidated retained earnings is a lot shorter. At this point, we include three components (a fourth component that deals with changes in ownership interests will be introduced in Chapter 7). In Chapter 6, we will also see how the three components are broad enough to incorporate the impact arising from equity accounting of associates and joint ventures. The three components are explained below:

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Legal entities' retained earnings that are attributable to owners of the parent

This first component comprises the parent's legal entity retained earnings and the parent's share of the post-acquisition retained earnings of the subsidiaries. The parent has a share of the legal entity retained earnings of the subsidiaries from acquisition date to the reporting date. By taking only the parent's share, we are excluding non-controlling interests' share. By adding parent's retained earnings with parent's share of post-acquisition retained earnings of the subsidiaries, we are also automatically eliminating intra-group transactions that are fully offset at the point of reporting.

For example, consider the following scenario:

P's retained earnings as at 31 December 20x6	$1,000,000
S's retained earnings as at 31 December 20x6	$ 500,000
S's pre-acquisition retained earnings	$ 300,000
P's ownership interests in S acquired on 1 July 20x4	90%
Dividend income from S recognized by P during post-acquisition period	$ 90,000
Dividend declared by S during post-acquisition period	$ 100,000

Assuming that these are the only transactions of relevance,

Consolidated retained earnings as at 31 December 20x6

= P's retained earnings as at 31 December 20x6 + P's share of S's post-acquisition retained earnings to 31 December 20x6
= $1,000,000 + [90% × ($500,000 − $300,000)]
= $1,000,000 + $180,000
= $1,180,000

The dividend income from S of $90,000 that is recognized in P's legal entity retained earnings will be offset by 90% of dividends declared by S that is included in the $180,000 above. The auto-elimination process makes the analytical check an efficient and focused way to determine the consolidated retained earnings.

However, there are some items that will not be automatically eliminated or re-enacted in the first component above. This brings us to our second and third components below.

Cumulative amortization or expensing of fair value differentials or impairment of goodwill arising from acquisition date

From earlier chapters, we have learnt that under- or over-valued identifiable net assets at acquisition date of a subsidiary would require consolidation adjustments to recognize higher or lower expensing for the acquirer. Higher or lower expenses arise when these under- or over-valued identifiable net assets are sold, depreciated or consumed by the acquiree. Because these adjustments are not pushed down to the subsidiary's legal entity books, they would not be

reflected in parent's share of post-acquisition retained earnings of the subsidiary as shown in the first element above. Hence, the analytical check of consolidated retained earnings will require recognition of the parent's share of the cumulative amortization or expensing of fair value differentials arising as at acquisition date. The same applies to the goodwill impairment loss that is attributable to the parent.

Let us extend our scenario above. Assume that S's intangible asset was undervalued by $50,000 at acquisition date and had a remaining useful life of ten years from acquisition date. Consolidated retained earnings would reflect the amortization of 2.5 years of the intangible asset during the post-acquisition period from 1 July 20x4 to 31 December 20x6. On an after-tax effect, P's share of the amortization would reduce consolidated retained earnings by $9,000 (90% x 80% × $50,000/10 × 2.5).

Hence, the consolidated retained earnings balance at 31 December 20x6 is:

P's retained earnings	$1,000,000	
P's share of post-acquisition retained earnings of S	180,000	Reflects the expensing that is not in S' books
P's share of cumulative amortization of the intangible asset, after-tax	(9,000)	

Remaining unrealized profit or losses from upstream or downstream transfers

Finally, we have to note that P's legal entity retained earnings may include remaining unrealized profit from downstream transfers, and P's share of post-acquisition retained earnings of S may include remaining unrealized profit from upstream transfers. These remaining unrealized profit or loss at reporting date must be removed to arrive at the consolidated retained earnings. For downstream transfers, the entire remaining amount has to be removed as P is not sharing the adjustment with non-controlling interests. For upstream transfers, P's share of the unrealized profit at reporting date has to be removed as non-controlling interests has a share of the adjustment as well. The amount removed for unrealized profit from upstream transfer is the same as the adjustment to arrive at non-controlling interests' balance. The after-tax effects should be shown as the adjustments.

Continuing with our illustration, let's assume the following additional information:

Transfer of inventory from P Co to S Co	
Transfer price:	$100,000
Original cost:	$ 70,000
% unsold at 31 December 20x6:	30%
Transfer of equipment from S Co to P Co	
Transfer price:	$120,000
Net book value at date of sale:	$100,000
Remaining useful life at date of sale:	10 years
Remaining useful life at 31 December 20x6:	6 years

With the additional information, the consolidated retained earnings balance at 31 December 20x6 is:

P's retained earnings	$1,000,000
P's share of post-acquisition retained earnings of S	180,000
P's share of cumulative amortization of the intangible asset, after-tax	(9,000)
P's share of unrealized profit from upstream sale, after-tax (Note A)	(8,640)
P's unrealized profit from downstream sale, after-tax (Note B)	(7,200)

In this illustration, we reduce consolidated retained earnings by $8,640 for the upstream sale because there is an originating profit in S Co's retained earnings that include the internal profit. We also remove $7,200, which is 100% of the unrealized profit from the downstream transfer.

Note A: P's share of unrealized profit from upstream sale
= 90% × 80% × ($20,000/10 × 6)
= $8,640

Note B: P's unrealized profit from downstream sale
= 30% × 80% × ($100,000 - $70,000)
= $7,200

In Chapter 7, we will introduce a fourth component arising from changes in ownership interests.

Figure 5.3 summarizes the components in consolidated retained earnings as of now.

FIGURE 5.3 Components of consolidated retained earnings

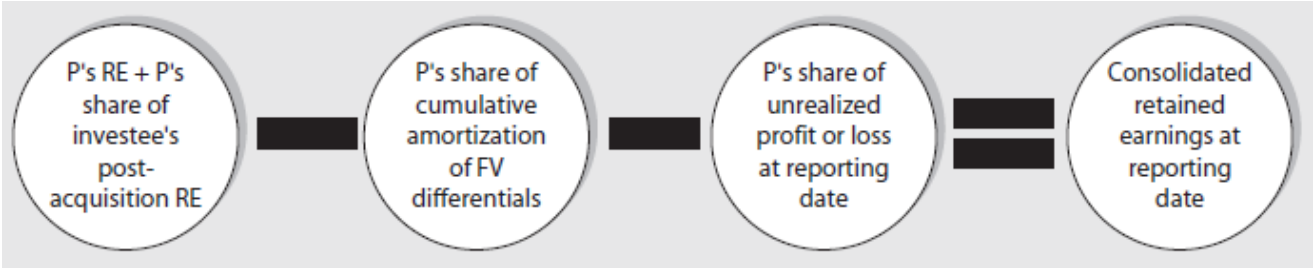


Illustration 5.4 provides a comprehensive example of accounting for goodwill, fair value adjustments, non-controlling interests, and intercompany transfers of fixed assets and inventory.

ILLUSTRATION 5.4 Comprehensive example with intercompany transfers

Income Statement and Partial Statement of Changes in Equity
For the Year Ended 31 December 20x5

	P Co	S Co
Operating profit	$1,000,000	$300,000
Dividend income (including from S)	42,000	

Profit before tax	$1,042,000	$300,000
Tax expense	(200,000)	(60,000)
Profit after tax	$ 842,000	$240,000
Dividends declared	(150,000)	(40,000)
Profit retained	$ 692,000	$200,000
Retained earnings, 1 January 20x5	900,000	500,000
Retained earnings, 31 December 20x5	$1,592,000	$700,000

Statement of Financial Position
As at 31 December 20x5

	P Co	S Co
Fixed assets, net book value	$ 900,000	$ 700,000
Investment in S, at cost	540,000	
Other investments	160,000	
Inventory	350,000	200,000
Accounts receivable	200,000	150,000
Cash	50,000	40,000
	$2,200,000	$1,090,000
Accounts payable	$ 258,000	$ 90,000
Share capital	350,000	300,000
Retained earnings	1,592,000	700,000
	$2,200,000	$1,090,000
Date of acquisition		1 July 19x9
Percentage acquired by P Co		90%
Shareholders' equity at date of acquisition:		
Share capital		$300,000
Retained earnings		100,000
		$400,000
Fair and book values of net assets at date of acquisition:		
Fixed assets		
Fair value		$160,000
Book value		120,000
Excess of fair value over book value		$ 40,000
Unrecognized intangible assets (fair value)		60,000

Other net assets (book and fair values)	280,000
Fair value of identifiable net assets at date of acquisition . . .	500,000
Book value of identifiable net assets at date of acquisition . .	400,000
Fair value of non-controlling interests at date of acquisition .	60,000

Additional information:

(a) The remaining useful life of fixed assets of S Co as at date of acquisition is five years with no residual value at the end of the useful life.

(b) The unrecognized intangible asset at acquisition date had an indefinite useful life and showed no indications of impairment.

(c) S Co transferred its fixed asset to P Co on 1 January 20x4. Details are as follows:

Transfer price invoiced by S Co to P Co		$30,000
Original cost of the fixed asset	$60,000	
Accumulated depreciation	(36,000)	
Net book value at 1 January 20x4		24,000
Profit on sale recorded by S Co in 20x4		$ 6,000
Useful life from 1 January 20x1, the original purchase date		5 years

There was no revision in the estimate of useful life (that is, remaining life at 1 January 20x4 was two years) and residual value was negligible.

(d) S Co purchased inventory from P Co on 1 December 20x4. Details are as follows:

Transfer price	$20,000
Original cost	15,000
Profit on sale recorded by P Co	$ 5,000
80% of the inventory was resold to third parties only in 20x5 with 20% remaining in S's inventory as at 31 December 20x5	

(e) S Co sold inventory to P Co during 20x5. Details are as follows:

Transfer price	$100,000
Original cost	80,000
Profit on sale recorded by S Co	$ 20,000
60% remained in P's inventory as at 31 December 20x5	

(f) Assume a tax rate of 20%. Tax on fair value adjustments should be recognized.

Required:

1. Prepare the consolidation entries for 20x5.
2. Perform the analytical check on the balance of non-controlling interests.

3. Determine the consolidated amounts of the following items:

   (a) Fixed assets balance as at 31 December 20x5.
   (b) Inventory balance as at 31 December 20x5.
   (c) Retained earnings balance as at 1 January 20x5.
   (d) Net profit after tax for the year ended 31 December 20x5.
   (e) Retained earnings balance as at 31 December 20x5.

1. Journal entries for 20x5

CJE1: Elimination of investment in S Co

Dr Share capital	300,000	
Dr Retained earnings	100,000	
Dr Fixed assets	40,000	
Dr Intangible asset	60,000	
Dr Goodwill (Note 2)	120,000	
Cr Investment in S		540,000
Cr Non-controlling interests (Note 3)		60,000
Cr Deferred tax liability (Note 1)		20,000

Note 1: Deferred tax liability = Tax rate × Fair value adjustments
= 20% × (Excess of fair value over book value of fixed assets + Fair value of intangible assets)
= 20% × ($40,000 + $60,000)
= 20% × $100,000
= $20,000

Note 2: Goodwill = Consideration transferred + Fair value of non-controlling interests – Fair value of identifiable net assets (after incorporating deferred tax liability)
= $540,000 + $60,000 – ($500,000 – $20,000)
= $600,000 – $480,000
= $120,000

Note 3: Non-controlling interests are at fair value at acquisition date. Fair value is proportionate to consideration transferred by P.

CJE2: Past depreciation of undervalued fixed assets (fully depreciated)

Dr Opening retained earnings	36,000	
Dr Non-controlling interests	4,000	
Cr Accumulated depreciation		40,000

CJE3: Tax effects of CJE2

Dr Deferred tax liability	8,000	

Cr Opening retained earnings		7,200
Cr Non-controlling interests		800

CJE4: Adjustment for unrealized profit on transfer of fixed assets

Dr Opening retained earnings	5,400	
Dr Non-controlling interests	600	
Dr Fixed assets	30,000	
Cr Accumulated depreciation		36,000

	What should be	What is	Adjustment
Fixed assets, cost	60,000	30,000	30,000
Accumulated depreciation	(36,000)	0	(36,000)
Net book value	24,000	30,000	6,000

CJE5: Adjustment for tax on unrealized profit on transfer of fixed assets

Dr Deferred tax asset	1,200	
Cr Opening retained earnings		1,080
Cr Non-controlling interests		120

CJE6: Adjustment of past and current depreciation of transferred fixed asset

Dr Accumulated depreciation	6,000	
Cr Opening retained earnings		2,700
Cr Non-controlling interests		300
Cr Depreciation		3,000
Depreciation before transfer		$12,000 ($60,000/5)
Depreciation after transfer		15,000 ($30,000/2)
Annual over-depreciation to be corrected		$ 3,000

Alternatively: Excess depreciation = Unrealized profit/Remaining useful life
= $6,000/2
= $3,000

CJE7: Tax effects of CJE6

Dr Opening retained earnings	540	
Dr Non-controlling interests	60	
Dr Tax expense	600	
Cr Deferred tax asset		1,200

CJE8: Adjustment for unrealized profit from downstream sale in beginning inventory

Dr Opening retained earnings	5,000	
Cr Cost of sales		4,000
Cr Inventory		1,000

CJE9: Tax effects of CJE8

Dr Tax expense	800	
Dr Deferred tax asset	200	
Cr Opening retained earnings		1,000

CJE10: Adjustment for unrealized profit from upstream sale during 20x5

Dr Sales	100,000	
Cr Cost of sales		88,000
Cr Inventory		12,000

CJE11: Tax effects of CJE10

Dr Deferred tax asset	2,400	
Cr Tax expense		2,400

CJE12: Allocate share of post-acquisition retained earnings to non-controlling interests

Dr Opening retained earnings	40,000	
Cr Non-controlling interests		40,000

Retained earnings as at 1 January 20x5	$500,000
Retained earnings as at date of acquisition	100,000
Change in retained earnings	$400,000
Non-controlling interests' share at 10%	40,000

CJE13: Eliminate dividends declared by S Co

Dr Dividend income	36,000	
Dr Non-controlling interests	4,000	
Cr Dividends declared		

CJE14: Allocate share of current income to non-controlling interests

Dr Income to non-controlling interests	23,280	
Cr Non-controlling interests		23,280

Net profit after tax of S Co	$240,000

Add excess depreciation on transferred fixed asset	3,000
Less tax on excess depreciation on transferred fixed asset	(600)
Less unrealized profit from upstream sale	(12,000)
Add tax on unrealized profit from upstream sale	2,400
Adjusted net profit after tax	$232,800
Non-controlling interests' share at 10%	23,280

2. Analytical check on non-controlling interests

Book value of net assets as at 31 December 20x5	$1,000,000
Balance of unrealized profit in inventory (after-tax)	(9,600)
	$ 990,400
Share of adjusted net assets	$ 99,040
Goodwill (Note 1)	12,000
Intangible asset (after-tax)	4,800
Non-controlling interests	$ 115,840
CJE1: Non-controlling interests at date of acquisition	$ 60,000
CJE2: Share of past depreciation of undervalued asset	(4,000)
CJE3: Tax effects of past depreciation of undervalued asset	800
CJE4: Share of adjustment of profit on transfer of fixed assets	(600)
CJE5: Share of tax on adjustment of profit on asset transfer	120
CJE6: Share of adjustment to past depreciation	300
CJE7: Share of tax on past depreciation	(60)
CJE12: Share of post-acquisition retained earnings	40,000
CJE13: Dividends received	(4,000)
CJE14: Allocate share of current income to non-controlling interests	23,280
Non-controlling interests' balance as at 31 December 20x5	$ 115,840

Note 1: Goodwill attributable to non-controlling interests (NCI)
= Fair value of NCI – Share of identifiable net assets
= $60,000 – (10% × $480,000)
= $12,000

3. Consolidated amounts

(a) Fixed assets balance as at 31 December 20x5

P Co	$ 900,000
S Co	700,000
Consolidated fixed assets balance as at 31 December 20x5	$1,600,000

The undervalued fixed asset has been fully depreciated. Hence, the remaining balance of the undervalued amount is zero. The unrealized profit on the transferred fixed asset is fully extinguished through depreciation. Hence, there is no need to remove the unrealized profit from the combined balance.

(b) Inventory balance as at 31 December 20x5

P Co	$350,000	
S Co	200,000	
	$550,000	
Less:		
Remaining unrealized profit on downstream transfer	(1,000)	20% × $5,000
Remaining unrealized profit on upstream transfer	(12,000)	60% × $20,000
Consolidated inventory balance as at 31 December 20x5	$537,000	

(c) Retained earnings balance as at 1 January 20x5

P Co's retained earnings	$ 900,000	
P Co's share of S Co's post-acquisition retained earnings	360,000	90% × ($500,000 - $100,000)
Past periods' depreciation of undervalued fixed assets (after-tax)	(28,800)	90% × 80% × $40,000
Remaining unrealized profit on fixed asset upstream transfer (after-tax)	(2,160)	90% × 80% × $6,000/2
Remaining unrealized profit on inventory downstream transfer (after-tax)	(4,000)	80% × $5,000
Consolidated retained earnings as at 1 January 20x5	$1,225,040	

As retained earnings is an equity item that is attributable to P Co's shareholders, adjustments for upstream transfers and depreciation of undervalued fixed assets is pro-rated by P Co's ownership interests in S Co. Since retained earnings is a fairly complex account, we present the compilation of consolidation adjustments to arrive at the same balance:

P Co	$ 900,000
S Co	500,000
	$1,400,000
Entries that impact opening retained earnings:	
CJE1	(100,000)
CJE2	(36,000)
CJE3	7,200
CJE4	(5,400)
CJE5	1,080
CJE6	2,700
CJE7	(540)

CJE8	(5,000)
CJE9	1,000
CJE12	(40,000)
Consolidated opening retained earnings	$1,225,040

(d) Profit after tax

P Co	$ 842,000	
S Co	240,000	
	$1,082,000	
Add excess depreciation (after-tax) from previous year's transfer of fixed assets	2,400	80% × $3,000
Add realized profit (after-tax) from previous year's transfer of inventory	3,200	80% × 80% × $5,000
Less unrealized profit (after-tax) from current year's transfer of inventory	(9,600)	80% × 60% × $20,000
Less dividend income eliminated	(36,000)	80% × $5,000
Consolidated profit after tax	$1,042,000	

All adjustments have to be on an after-tax basis. Unrealized profit from current year transfers are removed out of profit. Realized profit from previous year transfers are added back to profit. Both upstream and downstream transfers are treated in the same manner. Non-controlling interests' share is shown after the consolidated profit after tax line.

(e) Retained earnings as at 31 December 20x5

P Co's retained earnings	$1,592,000	
P Co's share of S Co's post-acquisition retained earnings	540,000	90% × ($700,000 - $100,000)
Less Cumulative depreciation of undervalued fixed assets (after-tax)	(28,800)	90% × 80% × $40,000
Less Remaining unrealized profit on inventory downstream transfer (after-tax)	(800)	20% × 80% × $5,000
Less Remaining unrealized profit on inventory upstream transfer (after-tax)	(8,640)	90% × 60% × 80% × $20,000
Consolidated retained earnings as at 31 December 20x5	$2,093,760	

ILLUSTRATION 5.4 EXTENSION Comprehensive example with intercompany transfers

The information in Illustration 5.4 remains the same except that P Co measures non-controlling interests as at acquisition date as a proportion of identifiable net assets.

CJE1: Elimination of investment in S Co

Dr Share capital	300,000	
Dr Retained earnings	100,000	
Dr Fixed assets	40,000	
Dr Intangible asset	60,000	
Dr Goodwill (Note 2)	108,000	
Cr Investment in S		540,000
Cr Non-controlling interests (Note 3)		48,000
Cr Deferred tax liability (Note 1)		20,000

The remaining consolidation entries will be the same as for Illustration 5.4. Goodwill attributable to non-controlling interests does not feature in Illustration 5.4 Extension. If there is no impairment of goodwill in Illustration 5.4, the allocation of current and past income to non-controlling interests remains the same. The analytical check on non-controlling interests excludes goodwill attributable to non-controlling interests.

Note 1: Deferred tax liability = Tax rate × Fair value adjustments
= 20% × (Excess of fair value over book value of fixed assets + Fair value of intangible assets)
= 20% × ($40,000 + $60,000)
= 20% × $100,000
= $20,000

Note 2: Goodwill = Consideration transferred + Fair value of non-controlling interests – Fair value of identifiable net assets (after incorporating deferred tax liability)
= $540,000 + $48,000 – ($500,000 – $20,000)
= $588,000 – $480,000
= $108,000

Note 3: Non-controlling interests are measured as a proportion of identifiable net assets.

Fair value of identifiable net assets (after incorporating deferred tax liability) is $480,000. Non-controlling interests as a proportion of identifiable net assets is $48,000 (10% × $480,000).

Analytical check on non-controlling interests:

Book value of net assets as at 31 December 20x5	$1,000,000
Balance of unrealized profit in inventory (after-tax)	(9,600)
	990,400
Intangible asset (after-tax)	48,000
Adjusted net assets as at 31 December 20x5	$1,038,400
Non-controlling interests' share at 10%	$ 103,840
(As fair value of non-controlling interests is proportionate to consideration transferred by P, goodwill is shared in the same proportion)	
CJE1: Non-controlling interests at date of acquisition	$ 48,000
CJE2: Share of past depreciation of undervalued asset	(4,000)

CJE3: Tax effects of past depreciation of undervalued asset	800
CJE4: Share of adjustment of profit on transfer of fixed assets	(600)
CJE5: Share of tax on adjustment of profit on asset transfer	120
CJE6: Share of adjustment to past depreciation	300
CJE7: Share of tax on past depreciation	(60)
CJE12: Share of post-acquisition retained earnings	40,000
CJE13: Dividends received	(4,000)
CJE14: Allocate share of current income to non-controlling interests	23,280
Non-controlling interests' balance as at 31 December 20x5	$ 103,840

SPECIAL ACCOUNTING CONSIDERATIONS WHEN INTRAGROUP TRANSFERS ARE MADE AT A LOSS

When intragroup transfers of inventory or fixed assets result in a loss to the selling company, IFRS 10 Appendix B paragraph B86(c) requires an assessment of whether the loss is indicative of an impairment loss that requires recognition in the consolidated financial statements.

For example, a parent may sell inventory below cost to a subsidiary because the inventory is slow-moving and one, which the parent is not able to sell in the open market. In such a situation, the loss on the sale of inventory may indicate an impairment loss that should be recognized in the consolidated financial statements. It will be inappropriate to remove the loss from the inventory. The consolidation adjustments will thus require the elimination of sale and cost of sales, without adjusting for the loss in the inventory. The same arguments apply to the transfer of fixed assets. When the transfer price is below the net book value and the loss indicates an impairment loss in the fixed assets, the unrealized loss should not be adjusted from the fixed asset. Illustration 5.5 explains the treatment of loss arising from intragroup transfers.

ILLUSTRATION 5.5 Loss arising from intragroup transfers

Parent Co transferred inventory to SubCo during the year ended 31 December 20x6. No sales of the inventory were made to third parties by SubCo during 20x6. The loss on transfer indicated an impairment loss on the asset. The inventory transfer is:

Transfer price	$ 60,000
Original cost	80,000
Gross loss	$(20,000)

To retain the loss in the consolidated financial statements, the following consolidation adjustments are necessary in the light of the entries in Parent Co's and SubCo's books. The entry in Parent Co's books is:

Dr Cost of sales	80,000

Cr Inventory		80,000
Dr Amount from SubCo	60,000	
Cr Sales		60,000

The entry in SubCo's books shows:

Dr Inventory	60,000	
Cr Amount from Parent Co		60,000

The consolidation adjustment is:

CJE1: Eliminate the transfer of inventory

Dr Sales	60,000	
Cr Cost of sales		60,000

In this scenario, we should leave the inventory at the transfer price of $60,000 because that price reflects the net realizable value of the inventory. Thus, no adjustment is made to remove the unrealized loss. The sales are eliminated in full while the cost of sales is credited only partially. By crediting $60,000 to cost of sales instead of $80,000, there is an implicit recognition of $20,000 of loss in the consolidated income statement.

The fixed asset transfer is:

Transfer price		$120,000
Original cost	$200,000	
Accumulated depreciation	50,000	
Net book value at date of transfer		150,000
Loss on transfer		$(30,000)

If the loss on transfer is indicative of an impairment loss, the loss should not be removed from fixed assets.

CJE2: Adjustment for the transfer of fixed assets

Dr Fixed assets	80,000	
Cr Accumulated depreciation		80,000

Reinstatement of cost to original cost = Original cost – Transfer price
= $200,000 – $120,000
= $80,000

Reinstatement of accumulated depreciation	$50,000
Recognition of impairment loss on fixed asset	30,000
Adjustment to accumulated depreciation	$80,000

In the consolidated income statement, the impairment loss of $30,000 is recognized by not eliminating the loss on transfer. A reclassification adjustment will be appropriate to reclassify the loss on sale to an impairment loss.

CJE3: Reclassification of loss on sale to impairment loss

Dr Impairment loss	30,000	
Cr Loss on sale		30,000

In the consolidated statement of financial position, the net book value of fixed assets is $120,000, which incorporates the impairment loss of $30,000 in the form of a higher accumulated depreciation.

Original cost	$200,000
Accumulated depreciation	(80,000)
Net book value	$120,000

Subsequent depreciation will take into account any revision in useful life as a result of the impairment in value.

Comparisons of Transfer Price, Carrying Amount and Fair Value

Illustration 5.5 presents a situation when the loss on transfer is wholly indicative of an impairment loss. However, a number of other situations exist when the loss on transfer is:

1. Wholly an artificial or "unrealized" loss. The transferor may choose to sell at an artificially low price to a group of company as a form of financial assistance or to capitalize on tax advantages, amongst other reasons.
2. A combination of artificial or unrealized loss and impairment loss.

To determine whether a loss on an intragroup transfer includes an impairment loss and/or artificial or unrealized loss, we have to compare the transfer price against the fair value of the asset at date of transfer and its carrying amount. Illustration 5.6 presents different situations whereby the loss on transfer of assets between group companies in each situation has a different economic substance as indicated by the transfer price, carrying amount, and fair value.

ILLUSTRATION 5.6 Transfers at a loss

In each of the following situations, Parent Co transferred inventory to Subsidiary Co on 4 April 20x1. For simplicity, we assume that in all the situations, the inventory had not yet been resold to third parties. However, the principle that applies to the adjustments below is the same if there were partial sales to third parties. If the inventory had been resold to third parties, any adjustment to ending inventory would be pro-rated by the percentage of the remaining inventory. The situations are *not* exhaustive, but they illustrate the underlying principles that would apply to other situations.

The underlying principle in IFRS 10 is that impairment loss should be recognized as an expense in the consolidated income statement and should not be "reversed" out. We determine impairment loss by comparing the carrying amount of the asset with the appropriate measures. In determining impairment loss for inventory, we compare the carrying amount with net realizable value (NRV). For other non-financial assets, we compare the carrying amount with the recoverable amount. IAS 36 *Impairment of Assets* defines recoverable amount as the higher of the fair value of the asset less cost to sell and the asset's value in use. For simplicity, we will refer to the appropriate benchmark measure as "fair value" in this illustration and in the end-of-chapter questions.

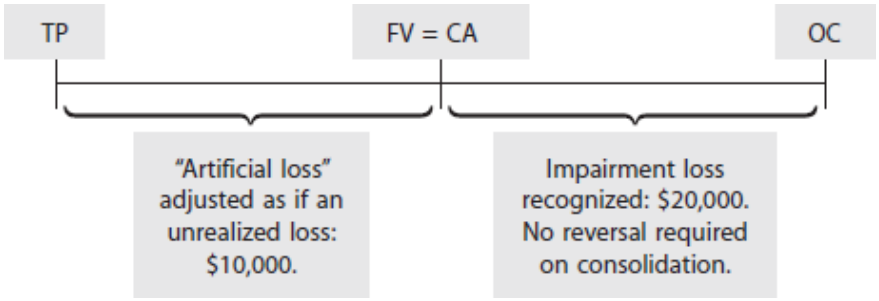
The underlying principle also requires loss on transfer that is not indicative of an impairment loss not to be recognized. In the same way that we do not recognize unrealized profit on sale in consolidated income because these are internal profits, we should also not recognize artificial or unrealized loss on sale between group companies. We

determine artificial loss by comparing the transfer price with the fair value of the asset. If the transfer is made at arm's-length pricing, the selling company would not incur any artificial loss.

In the following situations, we compare transfer price (TP), the carrying amount (CA) of the inventory, and fair value (FV) to determine the nature of the loss, and we show the consolidation adjustments that are required to eliminate and/or adjust the transactions. Since inventory is carried at the lower of cost and net realizable value (LCNRV), we show the original cost (OC) to show the extent of the impairment loss.

Situation A: In Situation A, the transfer price is lower than the fair value indicating that the loss is an artificial loss, which should be adjusted out. However, the fair value is lower than the original cost indicating that the asset is impaired. Parent Co recognizes a gross loss of $30,000, which is the difference between the transfer price and the original cost. An impairment loss of $20,000 is included in the total loss. In this situation, Parent Co (P) had recognized the impairment loss before the sale. (The LCNRV test is applied at end of each reporting period; it is not necessary for the holder of inventory to perform this test on a continuous basis. In this situation, if P had not written down the inventory to NRV, the loss would be recognized in any case in the income statement.) The artificial or unrealized loss of $10,000 has to be reversed.

Transfer price	$ 90,000
Original cost	120,000
Carrying amount in P's books	100,000
Fair value	100,000



The consolidated journal entry is:

Dr Sales	90,000	
Dr Inventory	10,000	
Cr Cost of sales		100,000
Eliminate intercompany transfer		

Inventory on consolidated statement of financial position:

Inventory on S's statement of financial position	$ 90,000
Consolidated journal entry (see above)	10,000
Inventory at LCNRV	$100,000

Gross loss in consolidated income statement:

Gross loss on P's PL	($30,000)
Consolidated journal entry (see above)	$10,000
Adjusted gross loss	($20,000)

Another way of analyzing the situation is to correct the year-end unrealized loss in the inventory. This is more intuitive and gets us to the same end result. In getting the balance sheet right, we would be able to make the correct adjustments to the income statement or retained earnings figure.

To get the balance sheet (or statement of financial position) right, we compare "what should be" reported in the group's balance sheet and "what is" reported in the legal entity's balance sheet. In a downstream transfer, the asset resides on the balance sheet of the subsidiary. The analysis is shown in the next box. We see that there is a difference of $10,000 between the two balance sheets. The original cost of the inventory for the group is $120,000 whereas the original cost for the buyer (the subsidiary) is based on the transfer price of $90,000. The LCNRV test shows a difference of $10,000 between the group and the legal entity's balance sheet.

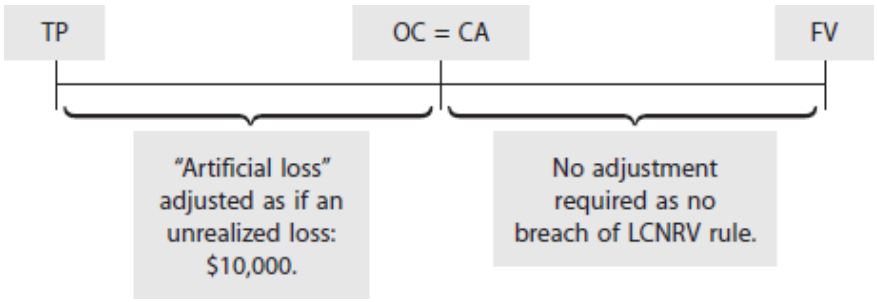
LCNRV test at year end	Group "What should be"	Legal entity "What is"	Difference
Original cost	$120,000	$ 90,000	
NRV	100,000	100,000	
LCNRV	100,000	90,000	$10,000
Impairment loss	20,000	0	$20,000

We see from the box above that additional impairment loss of $20,000 should be recognized on consolidation. The impairment loss is part of the cost of sales adjustment. To have a complete view of adjustments to cost of sales, the next box shows the comparison between the group and the legal entities cost of sales. It is not necessary to substantiate the cost of sales adjustment. If the sales and inventory adjustments are properly computed, cost of sales adjustment can be a residual. However, to facilitate a stronger understanding of the adjustment, we show the comparison below.

Income statement effect	Group "What should be"	Legal entity "What is"	Difference
Impairment loss	$20,000	$ 20,000 (recognized by P before the sale to S)	
Cost of sales ("sale" from P to S)	0	100,000 (after the write down to NRV)	
Total cost of sales	$20,000	$120,000	$100,000

Situation B: In Situation B, the transfer price is below the fair value indicating that the asset is not impaired. The loss on sale of $10,000 is wholly artificial and should be adjusted out from the income statement and inventory.

Transfer price	$ 90,000
Original cost	100,000
Fair value	120,000
Carrying amount in P's books	100,000



The consolidated journal entry is:

Dr Sales	90,000	
Dr Inventory	10,000	
Cr Cost of sales		100,000

Eliminate intercompany transfer

Inventory on consolidated statement of financial position:

As recorded by S	$ 90,000
Consolidated journal entry (see above)	10,000
LCNRV	$100,000

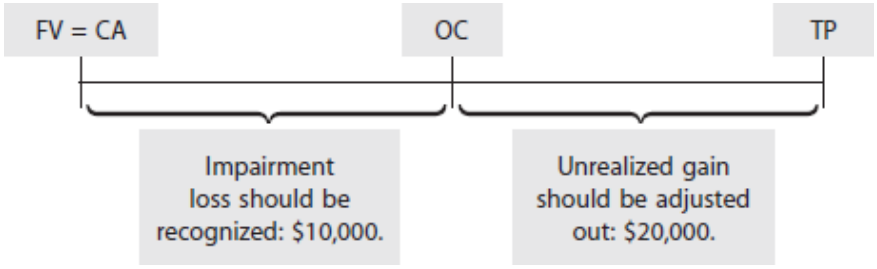
Gross loss in group's PL:

As recorded by P	($10,000)
Consolidated journal entry (see above)	10,000
Gross loss in group's PL	$ 0

LCNRV test at year end	Group "What should be"	Legal entity (Subsidiary Company) "What is"	Difference
Original cost	$100,000	$ 90,000	
NRV	120,000	120,000	
LCNRV	100,000	90,000	$10,000

Situation C:

Transfer price	$120,000
Original cost	100,000
Fair value	90,000
Carrying amount in P's books	90,000



LCNRV test at year end	Group	Legal entity (Subsidiary Company)	Difference
Original cost	$100,000	$120,000	
NRV	90,000	90,000	
LCNRV	90,000	90,000	$ 0
Impairment loss	10,000	30,000	20,000

The first consolidated journal entry is:

Dr Sales	120,000	
Cr Inventory		20,000
Cr Cost of sales		100,000
To reverse unrealized gain in inventory		

The second consolidated journal entry is:

Dr Inventory	20,000	
Cr Impairment loss (Cost of sales)		20,000
To adjust the excess impairment loss		

The combination of both journal entries is:

Dr Sales	120,000	
Cr Cost of sales		120,000
Eliminate intercompany transaction		

In this chapter, we analyzed the consolidation adjustments relating to intragroup transfers of inventory and fixed assets. However, the internal market place can be more complex. For example, one group company can be a contractor to another group company. IFRS 15 *Revenue from Contracts with Customers* as applied to the legal entity is inappropriate to the economic entity (an entity cannot be its contractor or customer). Other standards such as IAS 16 *Property, Plant and Equipment* applies instead to the economic entity. An illustrative example is provided in the Appendix to this chapter.

CONCLUSION

In this chapter, we learn a few key principles in the preparation of consolidated financial statements. Operational and financial synergies, economies of scale, and scope within the group are often generated through creating an internal market place. The internal market place is characterized by transactions and transfers between entities within the group. From an economic unit perspective, these transfers should not be recognized and internal profits should be eliminated. Consolidation principles require that assets and liabilities on the consolidated statement of financial position should not include profits or losses from intragroup transfers. These profits or losses are described as "unrealized profits or losses". Eventually, they are unwound or realized through the consumption, depreciation, expensing or sale of the transferred asset or liability. Intragroup balances should also be eliminated. Similarly, the consolidated income statement should reflect only expenses and revenues from external parties. Intragroup transactions should be eliminated in full. These include perfectly offsetting transactions during the period and those, which give rise to unrealized profits or losses. In this chapter, we note that accounting standards are applied at both group and legal entity levels, resulting in different recognition and measurement outcomes. The differences between the group and legal entity amounts are the subject of consolidation adjustments. We also see that special considerations relate to the accounting of losses that arise from intragroup transfers. In some situations, the losses should be recognized because the asset that is transferred is impaired. In other situations, the losses are artificial and should be reversed out in the same manner that a profit is reversed out. We also note that there are tax consequences from the elimination of intragroup profits and losses. Taxes incurred on intragroup profits or losses are deferred until the asset is expensed or sold by the buying group company.

In this chapter, we also evaluate the impact of profit adjustments on non-controlling interests. In upstream transfers, where the selling company is a partially owned subsidiary, the profit allocated to non-controlling interests would also be similarly adjusted for unrealized profits or losses. In downstream sales, where the selling company is the parent company, the profit allocated to non-controlling interests would not be affected.

This chapter also presents a new reconciliation that will enable us to arrive at consolidated totals expediently. This relates to the reconciliation or analysis of consolidated retained earnings. By deconstructing the consolidated retained earnings into three components, we are able to arrive at the final number independently of the consolidation adjustments.

APPENDIX 5A

Examples of Complex Intra-Group Transactions

Construction Activity within a Group

IFRS 15 *Revenue from Contracts with Customers* provides the principles and requirements relating to the timing, amount and disclosures of revenue. The transfer of control as evidenced by the accomplishment of separate performance obligation by the supplier to the customer is a central principle in IFRS 15 and is the pivot point for revenue recognition. The principles in IFRS 15 apply to transfer of control to the customer at a point in time as well as

continuous transfer of control to the customer over time. In Chapter 5, we established that transfer of control of assets must be made to third parties before the group is able to recognize the revenue.

In this Appendix, we apply the principles in IFRS 15 to the continuous transfer of control in an intragroup transaction. If one group company is a contractor in a construction project to its customer, another group company, the application of IFRS 15 has different results for the legal entities and the group entity. Figure 5A.1 shows how the contractor and customer recognizes the transaction as separate legal entities.

FIGURE 5A.1 Contractor and customer in the same group



The contractor recognizes revenue and expense as control is progressively transferred to the customer. Typically, the percentage of completion method is applied to recognize revenue and expense.[4] The customer recognizes fixed assets on the basis of progress billings. The billings are invoices sent to the customer to require payments throughout the construction period.

However, from the group's perspective, there is no transfer of control to third parties and revenue and expense should not be recognized. Instead, a self-constructed fixed asset comprising of construction costs should be recognized. In fact, the group should apply IAS 16 *Property, Plant and Equipment* and not IFRS 15. Figure 5A.2 shows how the self-constructed asset should be accounted.

FIGURE 5A.2 Self-constructed asset – "What Should Be"

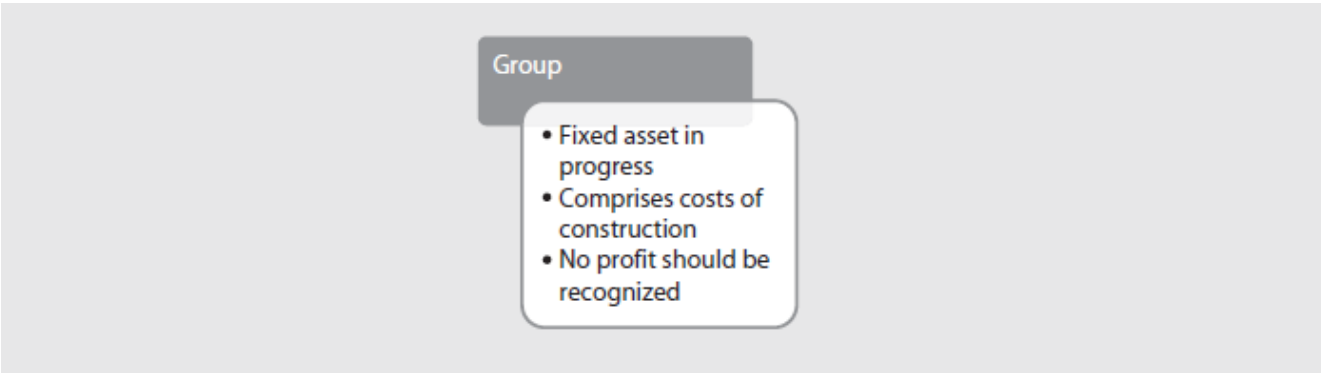


Illustration 5.7 presents a detailed illustration of how the contractor, customer and the group should account for the transaction.

ILLUSTRATION 5.7 Contract accounting within a group

Question

Co A undertakes construction work for its subsidiary as follows:

The following information relates to a fixed-price contract:

Contract price:	$50,000,000
Date of contract:	1 July 20x1

	31 Dec 20x1	31 Dec 20x2	31 Dec 20x3
Cumulative construction costs, 1 Jan	0	$9,000,000	$16,500,000
Current construction costs during year	$9,000,000	7,500,000	12,000,000
Estimated further costs to be incurred	36,000,000	27,500,000	19,000,000
Progress billings for the year	3,000,000	10,850,000	18,000,000
Cash received on progress billings	1,200,000	10,000,000	20,000,000

Assume that all construction costs are paid in cash.

Ignore tax effects.

Required:

1. Calculate the contract revenue, costs and gross profit under the percentage-of-completion method for the years ended 31 Dec 20x1, 31 Dec 20x2 and 31 Dec 20x3.
2. Prepare the journal entries to record all transactions relating to the construction contract for the years ended 31 Dec 20x1, 31 Dec 20x2 and 31 Dec 20x3 for the contractor.
3. Show the balances for the following as at 31 Dec 20x1, 31 Dec 20x2 and 31 Dec 20x3 for the contractor in its the Statement of Financial Position (extracts):
   (a) Contract Asset or Contract Liability
   (b) Accounts receivable
   (c) Cash
   (d) Retained earnings
4. Prepare the journal entries to record all transactions relating to the construction contract for the years ended 31 Dec 20x1, 31 Dec 20x2 and 31 Dec 20x3 for the customer.
5. Show the balances for the following as at 31 Dec 20x1, 31 Dec 20x2 and 31 Dec 20x3 for the customer in its Statement of Financial Position (extracts):
   (a) Fixed assets
   (b) Accounts payable
   (c) Cash or bank overdraft
7. Assuming that contractor is the parent and customer is the subsidiary, show the elimination entries on consolidation.
8. Show the balances for the following as at 31 Dec 20x1, 31 Dec 20x2 and 31 Dec 20x3 for the Group in its the Statement of Financial Position (extracts):
   (a) Fixed assets
   (b) Cash

Solution:

1. Contract revenue, costs and gross profit for 20x1, 20x2 and 20x3

Contract price: $50,000,000

Date of contract: 1 July 20x1

	31 Dec 20x1	31 Dec 20x2	31 Dec 20x3
Cumulative construction costs 1 Jan	0	$9,000,000	$16,500,000
Current construction costs during year	9,000,000	7,500,000	12,000,000
Estimated further costs to be incurred	36,000,000	27,500,000	19,000,000
Estimated total costs	45,000,000	44,000,000	47,500,000
Contract price	50,000,000	50,000,000	50,000,000
Estimated gross profit	5,000,000	6,000,000	2,500,000
Cumulative construction costs to 31 Dec	9,000,000	16,500,000	28,500,000
Percentage of completion	20%	37.50%	60.00%
Progress billings for the year	3,000,000	10,850,000	18,000,000
Amount paid on progress billings during year	1,200,000	10,000,000	20,000,000

	20x1	20x2	20x3
Cumulative revenue to 31 Dec	10,000,000	18,750,000	30,000,000
Cumulative revenue to 1 Jan	0	10,000,000	18,750,000
Current revenue	10,000,000	8,750,000	11,250,000
Cumulative costs to 31 Dec	9,000,000	16,500,000	28,500,000
Cumulative costs to 1 Jan	0	9,000,000	16,500,000
Current costs	9,000,000	7,500,000	12,000,000
Cumulative gross profit to 31 Dec	1,000,000	2,250,000	1,500,000
Cumulative gross profit to 1 Jan	0	1,000,000	2,250,000
Current gross profit/(loss)	1,000,000	1,250,000	(750,000)

2. Journal entries recorded by the Contractor

	20x1		20x2		20x3	
To recognize contract expense						
Dr Contract expense (PL).....	9,000,000		7,500,000		12,000,000	
Cr Cash.....		9,000,000		7,500,000		12,000,000
To recognize revenue						
Dr Contract asset (BS).....	10,000,000		8,750,000		11,250,000	
Cr Contract revenue (PL)....		10,000,000		8,750,000		11,250,000

Explanatory notes:

Contract asset represents the right to consideration for the contractor for the work done to date that is unbilled. Contract asset is eventually reduced when the progress billings are made to the customer. Under IFRS 15, there is no need to maintain a separate contra account for progress billings.

To record progress billings						
Dr Accounts receivable (BS) ...	3,000,000		10,850,000		18,000,000	
Cr Contract asset (BS).....		3,000,000		10,850,000		18,000,000

Explanatory notes:

Progress billings transform the asset from contract asset to accounts receivable. The accounts receivable represent unconditional payment expected from customer after passage of time.

To record cash received						
Dr Cash.....	1,200,000		10,000,000		20,000,000	
Cr Accounts receivable.....		1,200,000		10,000,000		20,000,000

3. Statement of Financial Position (extracts) of the Contractor

	31 Dec 20x1	31 Dec 20x2	31 Dec 20x3
Assets			
Contract asset	7,000,000	4,900,000	
Accounts receivable	1,800,000	2,650,000	650,000
Cash	(7,800,000)	(5,300,000)	2,700,000
Liabilities			
Contract liability			(1,850,000)
Net assets	1,000,000	2,250,000	1,500,000
Equity			
Retained earnings	1,000,000	2,250,000	1,500,000

4. Journal entries recorded by the Customer

	20x1		20x2		20x3	
To recognize progress billings (invoice from contractor) as fixed assets in progress						
Dr Fixed assets in progress.....	3,000,000		10,850,000		18,000,000	
Cr Accounts payable		3,000,000		10,850,000		18,000,000
To record cash paid						
Dr Accounts payable	1,200,000		10,000,000		20,000,000	
Cr Cash		1,200,000		10,000,000		20,000,000

5. Statement of Financial Position (extracts) for the Customer

	31 Dec 20x1	31 Dec 20x2	31 Dec 20x3
Assets			
Fixed assets in progress	3,000,000	13,850,000	31,850,000
Liabilities			
Bank overdraft	1,200,000	11,200,000	31,200,000
Accounts payable	1,800,000	2,650,000	650,000
	3,000,000	13,850,000	31,850,000

6. Consolidation adjustments if customer and contractor are within the same group

Explanatory notes:

We need to eliminate contract revenue, contract expense, contract assets or contract liabilities. Remember that an entity cannot have a contract account with itself. A "contractor" builds for "customers" and not for itself. Hence, the nature of the asset in the group should be "fixed assets", not "contract asset" or "contract liability". Next, we have to adjust the amount in fixed assets. The fixed assets should be the cumulative costs incurred to date, not the sum of progress billings.

		20x1		20x2		20x3	
CJE1: Eliminate contract revenue and expense							
Dr Opening RE.....	Note 1			1,000,000		2,250,000	
Dr Fixed assets.....	Note 2	6,000,000		2,650,000			
Dr Contract revenue	Note 3	10,000,000		8,750,000		11,250,000	
Dr Contract liability	Note 4					1,850,000	
Cr Fixed assets	Note 2						3,350,000
Cr Contract asset.....	Note 4		7,000,000		4,900,000		
Cr Contract expense	Note 5		9,000,000		7,500,000		12,000,000
		16,000,000	16,000,000	12,400,000	12,400,000	15,350,000	15,350,000

Note 1: Remove the cumulative gross profit included in the opening retained earnings (RE)

Note 2: Adjust the difference between fixed assets at two levels

Fixed assets (Group's SFP)	9,000,000	16,500,000	28,500,000
Fixed assets (Customer's SFP)	3,000,000	13,850,000	31,850,000
	6,000,000	2,650,000	(3,350,000)

Note 3: Eliminate the contract revenue for each period

Note 4: Eliminate the contract asset or contract liability at the end of each period (you can't have a contract with yourself)

Note 5: Eliminate the contract expense for each period

	20x1		20x2		20x3	
CJE2: Eliminate intercompany balances						
Dr Accounts payable	1,800,000		2,650,000		650,000	
Cr Accounts receivable		1,800,000		2,650,000		650,000

Explanatory notes:

This entry is to eliminate intercompany balances from the contract. Note however, that this assumes that there are no other transactions that affect the intercompany balances.

7. Statement of Financial Position (extracts) for the Group

	31 Dec 20x1	31 Dec 20x2	31 Dec 20x3
Assets			
Fixed assets in progress	9,000,000	16,500,000	28,500,000
Liabilities			
Bank overdraft	9,000,000	16,500,000	28,500,000

Explanatory notes:

The Group reports only fixed assets in progress at cost. Bank overdraft increases by the same amount to show the cash paid on the costs incurred on the fixed assets. The simplifying assumption here is that the expenditures are incurred through cash payment. In a more realistic setting, the expenditures are incurred through credit purchases, use of own inventory and own fixed assets. We also assume that this is the only transaction for the group resulting in a bank overdraft.

CONCEPT QUESTIONS

State True or False and the reasons for your response. For multiple-choice questions, give the most appropriate answer.

CQ5.1 Payables and receivables in the group statement of financial position include balances with associate companies.

CQ5.2 Dividend income in the profit or loss of a parent company should be eliminated against dividend payable in the statement of financial position of a subsidiary company.

CQ5.3 A profit from an intercompany sale would result in either a higher cost of sales in the income statement or a higher inventory balance in the statement of financial position of the buying company.

CQ5.4 An adjustment needs to be made to the unrealized profit included in the inventory balance to avoid the understatement of group profits and the overstatement of group assets.

CQ5.5 Revenue is recognized in group accounts only when the control of inventory are transferred to a third party.

CQ5.6 The parent's theory underlies the procedure for reducing non-controlling interests' share of current profit with their share of unrealized profit arising from an upstream sale.

CQ5.7 In a downstream sale, when a parent sells to its subsidiary, non-controlling interests' share of current profit is reduced by the latter's share of the unrealized profit.

CQ5.8 If an item purchased at a marked-up price from a group company in the previous year remains unsold at the end of the current year, group inventory and gross profit will have to be reduced.

CQ5.9 If an item purchased at a marked-up price from a group company in the previous year is resold in the current year, group gross profit will increase and group beginning retained earnings will reduce.

CQ5.10 If a subsidiary sells equipment to its parent and recognizes a loss on sale in the previous year and the loss is not indicative of an impairment loss, which of the following is true?

(a) A consolidation adjustment to reduce retained earnings and fixed assets is required.
(b) A consolidation adjustment to reduce gain on sale and fixed assets is required.
(c) A consolidation adjustment to increase retained earnings and fixed assets is required.

CQ5.11 Assuming the situation in CQ5.10, group depreciation expense in each subsequent year of sale will

(a) Not be subject to any adjustment.
(b) Be the sum of the parent's and subsidiary's depreciation adjusted downwards for the depreciation.
(c) Be the sum of the parent's and subsidiary's depreciation adjusted upwards for the depreciation.

CQ5.12 If a subsidiary sells equipment to its parent in the current year at a loss because the value of the equipment is impaired, which of the following is true?

(a) A consolidation adjustment to reverse loss on sale and increase fixed assets is required.
(b) A consolidation adjustment to reverse loss on sale and decrease fixed assets is required.
(c) Loss on sale is reclassified as an impairment loss.

PROBLEMS

P5.1 Consolidation adjustments and worksheets

The trial balances for FIRE and WALL as of 31 December 20x3 are as follows:

	FIRE		WALL	
	Debit	Credit	Debit	Credit
Cash	$ 19,800		$ 3,200	
Accounts receivable	32,000		38,000	
Inventory	330,000		170,000	
Land	160,000		80,000	
Buildings and equipment	680,000		520,000	
Investment in WALL, at cost	196,000			
Cost of goods sold	372,000		159,600	
Depreciation	40,000		30,000	
Interest expense	32,000		10,400	
Other expenses (including tax)	44,850		52,000	
Dividends declared	60,000		30,000	
Accumulated depreciation		$ 280,000		$ 160,000
Accounts payable		184,800		70,000
Bonds payable		400,000		300,000
Bond premium				3,200
Share capital		240,000		160,000
Retained earnings		241,600		100,000
Sales		520,000		250,000
Other income		79,250		50,000
Dividend income from subsidiary		21,000		
Total	$1,966,650	$1,966,650	$1,093,200	$1,093,200

Additional information:

(a) On 1 January 20x2, FIRE purchased 70% of WALL's ordinary shares. At the date of acquisition, share capital of WALL was $160,000 and retained earnings were $40,000.

(i) The excess of the acquisition price over the underlying book value was assigned to:

- Buildings that had a fair value of $40,000 greater than book value and remaining useful life of ten years from 1 January 20x2; and
- Goodwill. Goodwill impairment amounting to 20% of its original cost was recognized in 20x2.

(ii) Fair value of WALL as at acquisition date was $280,000. Non-controlling interests as at 1 January 20x2 had a proportionate share in the fair value of WALL as at that date.

(b) During 20x2, the following transactions arose:

(i) WALL purchased inventory for $64,000 and sold it to FIRE for $96,000. FIRE resold $54,000 of the inventory during 20x3, while the balance of $42,000 remained unsold as at 31 December 20x3.

(ii) FIRE sold the building that it originally purchased for $40,000 to WALL for $64,000. Accumulated depreciation at the date of sale was $8,000. Estimated useful life at date of original purchase was ten years and at the date of resale was eight years. WALL had recognized a full year's depreciation on the building in 20x2.

(c) During 20x3, FIRE sold inventory purchased for $120,000 to WALL for $180,000. WALL resold 40% of the inventory during 20x3.

(d) Tax rate was 20%. Recognize tax effects where appropriate.

Required:

1. Show all consolidation adjustment and elimination entries for the year ended 31 December 20x3.
2. Perform an analytical check on the non-controlling interests in WALL as at 31 December 20x3.
3. Prepare the consolidation worksheets for FIRE and WALL for the year ended 31 December 20x3.
4. Determine the following amounts for 20x3 analytically and compare with the amounts in your consolidation worksheets in 3:
   (a) Consolidated cost of sales
   (b) Consolidated depreciation expense
   (c) Consolidated inventory balance
   (d) Consolidated carrying amount of buildings and equipment
   (e) Consolidated retained earnings

P5.2 Consolidation adjustments and elimination entries

The financial statements of Jewel Ltd and its subsidiary Opal Ltd are shown below:

Income Statement and Partial Statement of Changes in Equity
For the Year Ended 31 December 20x2

	Jewel Ltd	Opal Ltd
Sales	$10,000,000	$3,300,000
Operating profit	$ 1,000,000	$ 160,000
Dividend income from Opal	23,400	
Profit before tax	$ 1,023,400	$ 160,000
Tax	(220,000)	(35,200)
Profit after tax	$ 803,400	$ 124,800
Retained earnings, 1 January 20x2	2,050,000	210,000
Dividends declared	(100,000)	(29,250)
Retained earnings, 31 December 20x2	$ 2,753,400	$ 305,550

Statement of Financial Position
As at 31 December 20x2

	Jewel Ltd	Opal Ltd
Share capital	$2,350,000	$300,000
Retained earnings	2,753,400	305,550
Shareholders' equity	$5,103,400	$605,550
Investment in Opal, cost	$ 450,000	

Other net assets	4,653,400	$605,550
Net assets	$5,103,400	$605,550

Additional information:

(a) Jewel Ltd acquired the interest in Opal Ltd on 1 January 20x0 when the shareholders' equity of Opal was as follows:

Share capital	$300,000
Retained earnings	120,000
Shareholders' equity	420,000
Purchase consideration paid by Jewel Ltd	450,000
Percentage ownership by Jewel Ltd in Opal Ltd	80%

Fair value of identifiable net assets was close to the book value at acquisition date.

(b) Goodwill and non-controlling interests

Non-controlling interests are recognized at fair value on acquisition date.

Fair value of non-controlling interests as at 1 January 20x0 . . . . . $112,500

Goodwill impairment is as follows: $50,000 in previous years and $40,000 in the current year.

(c) Intercompany sales

Intercompany sales made by Jewel to Opal during 20x2	$100,000
Profit on intercompany sales included in Opal's inventory as at 31 December 20x2	30,000

Tax expense on profit on intercompany sales has been properly charged to the income statement of Jewel during 20x2.

(d) Taxation rate was 20%. Tax effects are to be considered wherever appropriate.

Required:

1. Prepare consolidation adjustment and elimination entries for the year ended 31 December 20x2. Show all relevant workings. Worksheets are not required.
2. Perform an analytical check on non-controlling interests as at 31 December 20x2.
3. Perform an analytical check on consolidated retained earnings as at 31 December 20x2.

P5.3 Consolidation and analytical check on non-controlling interests

Prism Co acquired 80% of the stock of Sapphire Co for $300,000 on 1 January 20x7. At acquisition date, Sapphire reported retained earnings of $150,000. The excess of Prism Co's acquisition cost over its share of Sapphire's book value was assigned to buildings and equipment that had a remaining life of ten years at acquisition date and deferred tax liability on the undervalued building and equipment.

The purchase consideration paid by Prism Co was proportional to Prism's share of the fair value of Sapphire Co as an entity. Non-controlling interests are to be measured at its share of fair value of Sapphire Co as at acquisition date. The financial statements of the two companies for the year ended 31 December 20x9 are shown below. Investment in Sapphire Co was carried at cost.

Statements of Financial Position
As at 31 December 20x9

	Prism	Sapphire
Sales	$1,000,000	$ 480,000
Cost of goods sold	(640,000)	(320,000)
Dividend income	16,000	0
Depreciation expense	(100,000)	(10,000)
Interest expense	(72,000)	(14,000)
Tax and other expenses	(44,000)	(76,000)
Profit retained	$ 160,000	$ 60,000
Retained earnings, 1 January	580,000	300,000
Dividends declared	(40,000)	(20,000)
Retained earnings, 31 December	$ 700,000	$ 340,000
Cash and receivables	$ 620,000	$ 420,000
Inventory	640,000	270,000
Land	260,000	150,000
Buildings and equipment, cost	1,500,000	200,000
Investment in Sapphire, cost	300,000	
Debits	$3,320,000	$1,040,000
Accumulated depreciation	$1,000,000	$ 80,000
Payables	1,220,000	420,000
Share capital	400,000	200,000
Retained earnings	700,000	340,000
Credits	$3,320,000	$1,040,000

On 1 January 20x9, Prism Co held inventory purchased from Sapphire Co during 20x8 for $15,000, which had been manufactured by Sapphire at a cost of $10,000. During 20x9, Sapphire sold goods costing $40,000 to Prism Co for $60,000. Prism sold the inventory on hand at the beginning of the year, but continued to hold 40% of its 20x9 purchases from Sapphire on 31 December 20x9. Tax rate was 20%.

Required:

1. Prepare all necessary consolidation elimination and adjustment entries for the year ended 31 December 20x9.
2. Prepare the consolidation worksheets for the year ended 31 December 20x9.
3. Perform an analytical check on the non-controlling interests' balance as at 31 December 20x9.
4. Determine the following consolidated amounts as at 31 December 20x9 analytically and compare with the balances in your consolidation worksheets in Part 3:

   (a) Inventory

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(b) Buildings and equipment, net of accumulated depreciation

(c) Retained earnings

P5.4 Intercompany transfer of fixed assets

Parent Co purchased an equipment on 1 January 20x1 for $200,000. The estimated useful life at that date was ten years with a nil residual value. On 1 July 20x5, Parent Co sold the equipment to Subsidiary Co for $120,000. Parent Co had a 90% ownership interest in Subsidiary Co. On 1 July 20x5, the remaining economic useful life was re-estimated as seven years. Net profit after tax of Subsidiary Co for 20x5 was $500,000. Assume a tax rate of 20% throughout.

Required:

1. Show the consolidated journal entries for 20x5 relating to the above transaction.
2. If the situation had been reversed in that it was Subsidiary Co that sold the equipment to Parent Co, show the consolidated journal entries for 20x5.

P5.5 Consolidation and analytical check on non-controlling interests

The financial statements of P Co and Y Co for the year ended 31 December 20x5 are shown below. P Co acquired a 90% interest in Y Co on 1 January 20x3.

Income Statement and Partial Statement of Changes in Equity
For the Year Ended 31 December 20x5

	P Co	Y Co
Profit before tax (including dividend income)	$2,500,000	$ 900,000
Tax	(500,000)	(180,000)
Profit after tax	$2,000,000	$ 720,000
Dividends declared	(300,000)	(140,000)
Profit retained	$1,700,000	$ 580,000
Retained earnings, 1 January 20x5	900,000	800,000
Retained earnings, 31 December 20x5	$2,600,000	$1,380,000

Statement of Financial Position
As at 31 December 20x5

	P Co	Y Co
Fixed assets, net book value	$2,500,000	$1,250,000
Investment in Y Co, at cost	1,200,000	
Other investments	300,000	
Inventory	750,000	500,000
Accounts receivable	420,000	150,000
Cash	50,000	100,000
	$5,220,000	$2,000,000